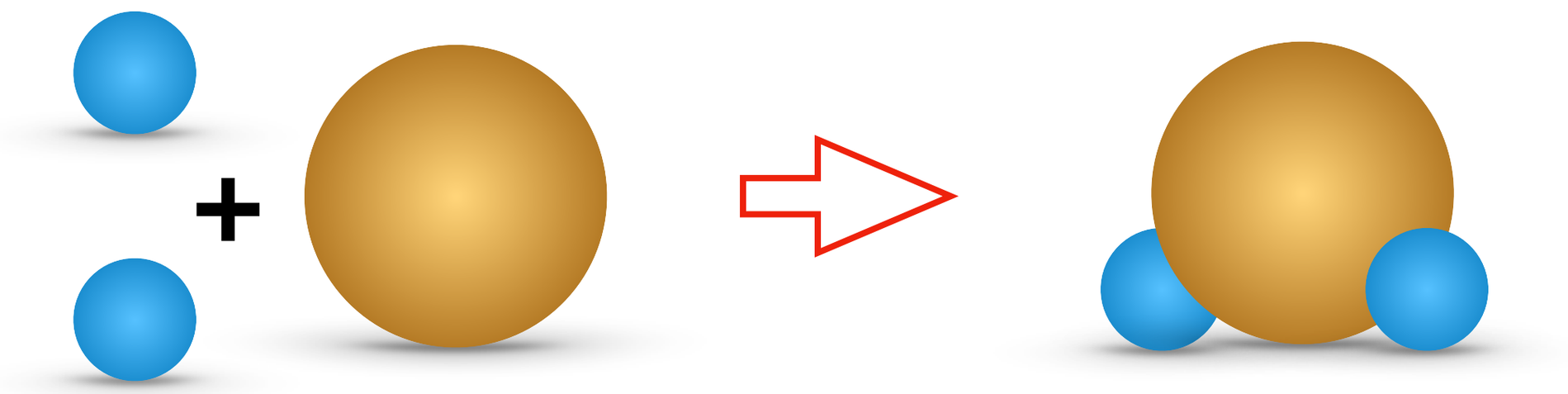
Fast and Small

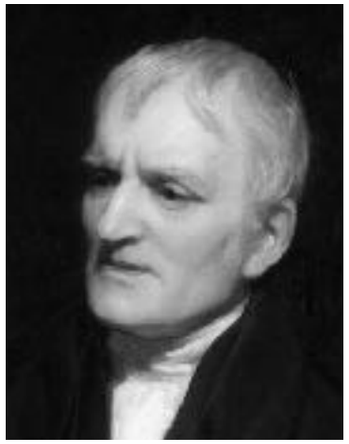
The Elementary World

Elements

Then

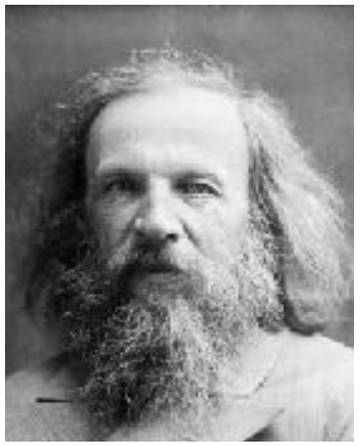
Elements





John Dalton

Reihen	Gruppo I. — R^2O	Gruppo II. — RO	Gruppo III. — R^2O^3	Gruppo IV. RH^4 RO^2	Gruppo V. RH^3 R^2O^5	Gruppo VI. RH^2 RO^3	Gruppo VII. RH R^2O^7	Gruppo VIII. — RO^4
1	H=1							
2	Li=7	Be=9,4	B=11	C=12	N=14	O=16	F=19	
3	Na=23	Mg=24	Al=27,3	Si=28	P=31	S=32	Cl=35,5	
4	K=39	Ca=40	—=44	Ti=48	V=51	Cr=52	Mn=55	Fe=56, Co=59, Ni=59, Cu=63.
5	(Cu=63)	Zn=65	—=68	—=72	As=75	Se=78	Br=80	
6	Rb=85	Sr=87	?Yt=88	Zr=90	Nb=94	Mo=96	—=100	Ru=104, Rh=104, Pd=106, Ag=108.
7	(Ag=108)	Cd=112	In=113	Sn=118	Sb=122	Te=125	J=127	
8	Cs=133	Ba=137	?Di=138	?Ce=140	—	—	—	— — — —
9	(—)	—	—	—	—	—	—	
10	—	—	?Er=178	?La=180	Ta=182	W=184	—	Os=195, Ir=197, Pt=198, Au=199.
11	(Au=199)	Hg=200	Tl=204	Pb=207	Bi=208	—	—	
12	—	—	—	Th=231	—	U=240	—	— — — —



Dmitri Mendeleev

Elements

Now

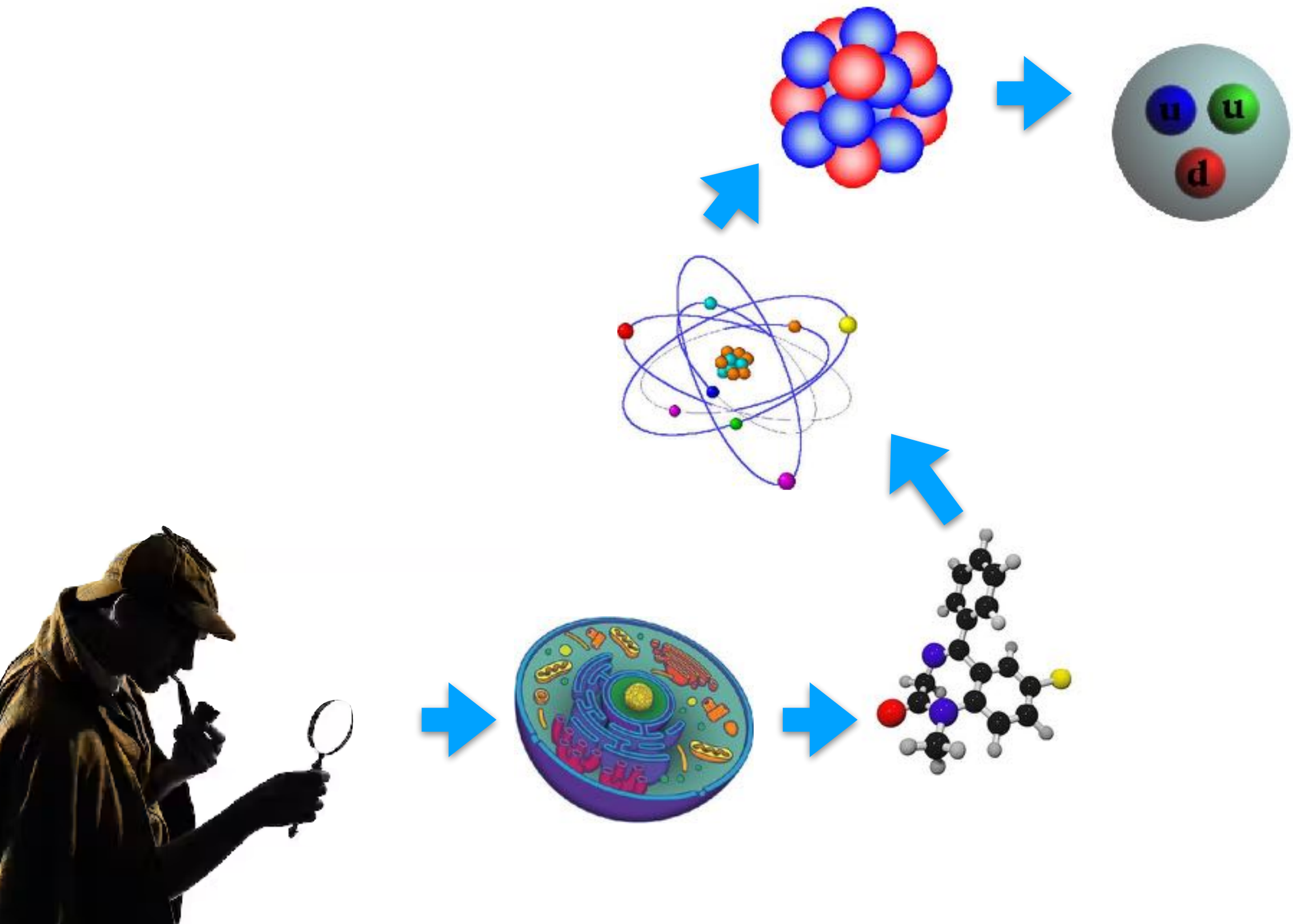
Periodic Table of the Elements

1 1IA 1A	2 IIA 2A	3 IIIB 3B	4 IVB 4B	5 VB 5B	6 VIB 6B	7 VIIB 7B	8 VIII 8	9 VIII 8	10 VIII 8	11 IB 1B	12 IIB 2B	13 IIIA 3A	14 IVA 4A	15 VA 5A	16 VIA 6A	17 VIIA 7A	18 VIIIA 8A
1 H Hydrogen 1.008																	2 He Helium 4.003
3 Li Lithium 6.941	4 Be Beryllium 9.012											5 B Boron 10.811	6 C Carbon 12.011	7 N Nitrogen 14.007	8 O Oxygen 15.999	9 F Fluorine 18.998	10 Ne Neon 20.180
11 Na Sodium 22.99	12 Mg Magnesium 24.305											13 Al Aluminum 26.982	14 Si Silicon 28.086	15 P Phosphorus 30.974	16 S Sulfur 32.066	17 Cl Chlorine 35.453	18 Ar Argon 39.948
19 K Potassium 39.098	20 Ca Calcium 40.078	21 Sc Scandium 44.956	22 Ti Titanium 47.867	23 V Vanadium 50.942	24 Cr Chromium 51.996	25 Mn Manganese 54.938	26 Fe Iron 55.845	27 Co Cobalt 58.933	28 Ni Nickel 58.693	29 Cu Copper 63.546	30 Zn Zinc 65.38	31 Ga Gallium 69.723	32 Ge Germanium 72.631	33 As Arsenic 74.922	34 Se Selenium 78.971	35 Br Bromine 79.904	36 Kr Krypton 83.789
37 Rb Rubidium 85.468	38 Sr Strontium 87.62	39 Y Yttrium 88.906	40 Zr Zirconium 91.224	41 Nb Niobium 92.906	42 Mo Molybdenum 95.95	43 Tc Technetium 98.907	44 Ru Ruthenium 101.07	45 Rh Rhodium 102.906	46 Pd Palladium 106.42	47 Ag Silver 107.868	48 Cd Cadmium 112.414	49 In Indium 114.818	50 Sn Tin 118.711	51 Sb Antimony 121.760	52 Te Tellurium 127.6	53 I Iodine 126.904	54 Xe Xenon 131.294
55 Cs Cesium 132.905	56 Ba Barium 137.328	57-71	72 Hf Hafnium 178.49	73 Ta Tantalum 180.948	74 W Tungsten 183.84	75 Re Rhenium 186.207	76 Os Osmium 190.23	77 Ir Iridium 192.217	78 Pt Platinum 195.085	79 Au Gold 196.967	80 Hg Mercury 200.592	81 Tl Thallium 204.383	82 Pb Lead 207.2	83 Bi Bismuth 208.980	84 Po Polonium [208.982]	85 At Astatine 209.987	86 Rn Radon 222.018
87 Fr Francium 223.020	88 Ra Radium 226.025	89-103	104 Rf Rutherfordium [261]	105 Db Dubnium [262]	106 Sg Seaborgium [266]	107 Bh Bohrium [264]	108 Hs Hassium [269]	109 Mt Meitnerium [278]	110 Ds Darmstadtium [281]	111 Rg Roentgenium [280]	112 Cn Copernicium [285]	113 Nh Nihonium [286]	114 Fl Flerovium [289]	115 Mc Moscovium [288]	116 Lv Livermorium [293]	117 Ts Tennessine [294]	118 Og Oganesson [294]

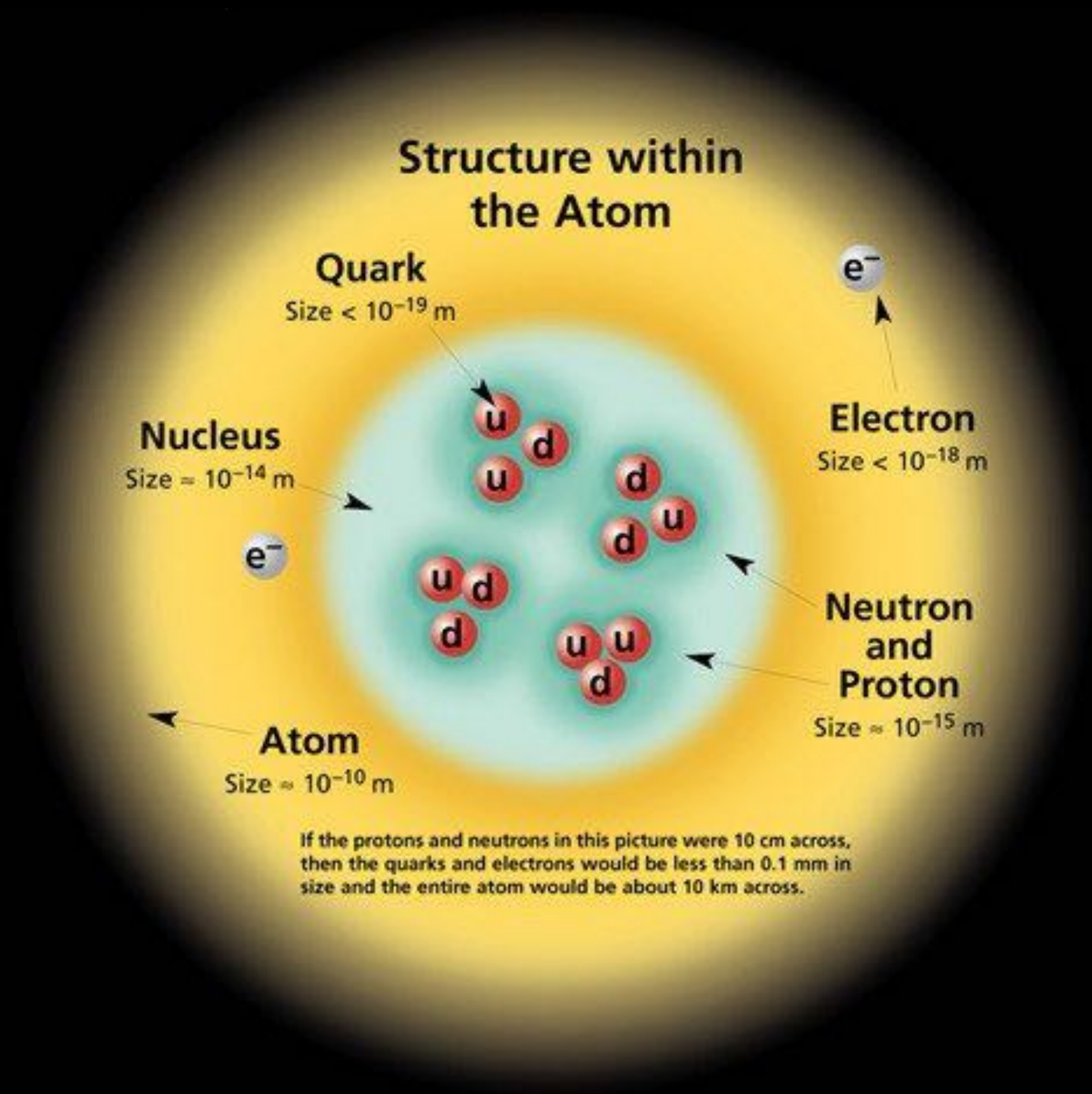
Lanthanide Series	57 La Lanthanum 138.906	58 Ce Cerium 140.116	59 Pr Praseodymium 140.908	60 Nd Neodymium 144.243	61 Pm Promethium 144.913	62 Sm Samarium 150.36	63 Eu Europium 151.964	64 Gd Gadolinium 157.25	65 Tb Terbium 158.926	66 Dy Dysprosium 162.500	67 Ho Holmium 164.930	68 Er Erbium 167.259	69 Tm Thulium 168.934	70 Yb Ytterbium 173.055	71 Lu Lutetium 174.967
Actinide Series	89 Ac Actinium 227.028	90 Th Thorium 232.038	91 Pa Protactinium 231.036	92 U Uranium 238.029	93 Np Neptunium 237.048	94 Pu Plutonium 244.064	95 Am Americium 243.061	96 Cm Curium 247.070	97 Bk Berkelium 247.070	98 Cf Californium 251.080	99 Es Einsteinium [254]	100 Fm Fermium 257.095	101 Md Mendelevium 258.1	102 No Nobelium 259.101	103 Lr Lawrencium [262]

Alkali Metal | Alkaline Earth | Transition Metal | Basic Metal | Semimetal | Nonmetal | Halogen | Noble Gas | Lanthanide | Actinide

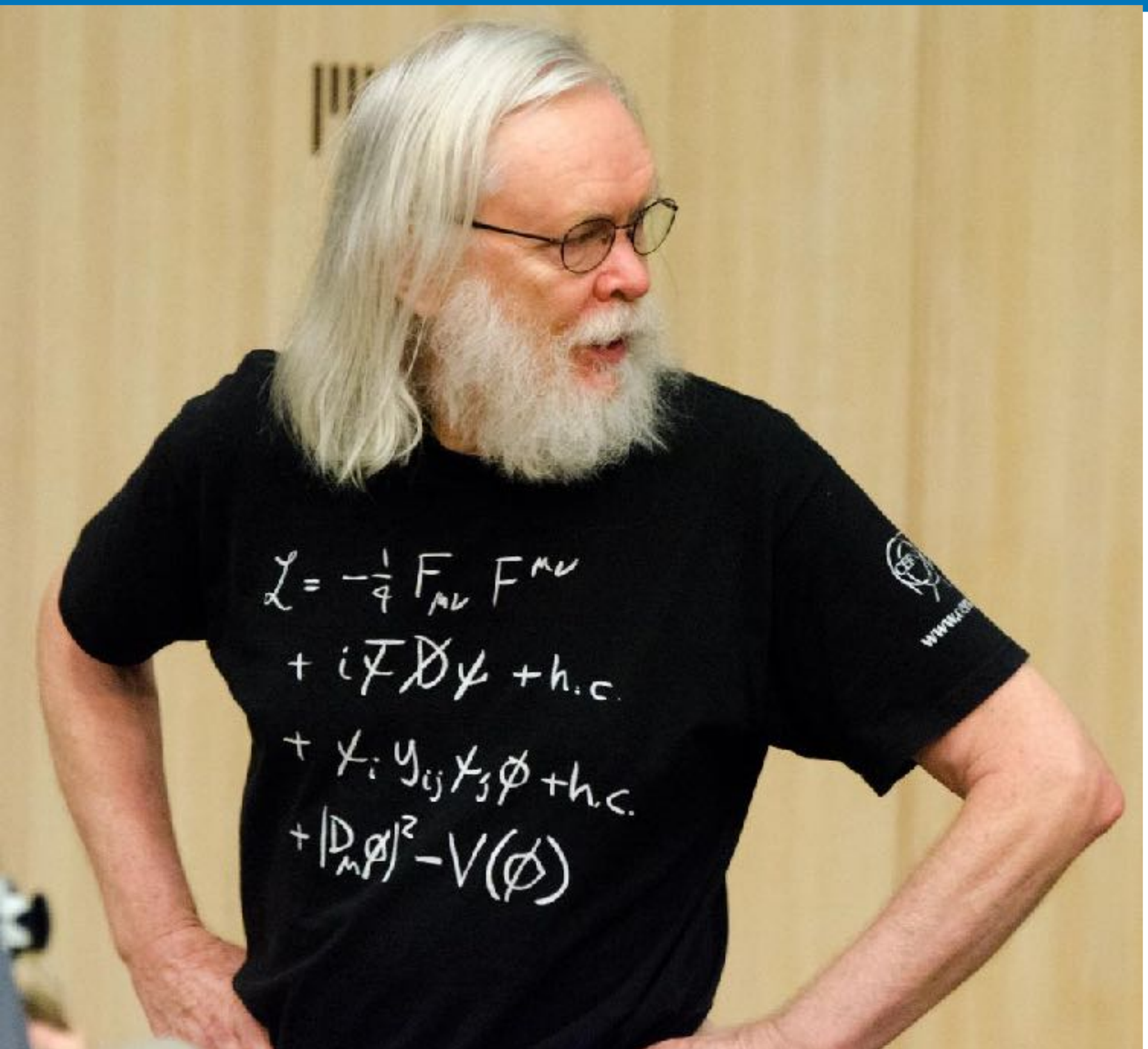
Elementary, … Watson!



Modern View


Structure within the Atom
Quark
Size < 10^{-19} m
Nucleus
Size ≈ 10^{-14} m
Electron
Size < 10^{-18} m
Neutron and Proton
Size ≈ 10^{-15} m
Atom
Size ≈ 10^{-10} m
e^-
u
d
If the protons and neutrons in this picture were 10 cm across, then the quarks and electrons would be less than 0.1 mm in size and the entire atom would be about 10 km across.

Modern View



The
Standard
Model

Modern View

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Details

Science
Volume 360, Issue 6385
Apr 2018

ARTICLE

Measurement of the fine-structure constant as a test of the Standard Model

View article page

Richard H. Parker, Chenghui Yu, Weicheng Zhong, Brian Estey and Holger Müller

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https://doi.org/10.1126/science.aap7706

Publisher	American Association for the Advancement of Science
ISSN	0036-8075
eISSN	1095-9203
Print	April 13, 2018

RESEARCH

METROLOGY

Measurement of the fine-structure constant as a test of the Standard Model

Richard H. Parker,[1*] Chenghui Yu,[1*] Weicheng Zhong,[1] Brian Estey,[1] Holger Müller[1,2]†

Measurements of the fine-structure constant α require methods from across subfields and are thus powerful tests of the consistency of theory and experiment in physics. Using the recoil frequency of cesium-133 atoms in a matter-wave interferometer, we recorded the most accurate measurement of the fine-structure constant to date: $\alpha = 1/137.035999046(27)$ at 2.0×10^{-10} accuracy. Using multiphoton interactions (Bragg diffraction and Bloch oscillations), we demonstrate the largest phase (12 million radians) of any Ramsey-Bordé interferometer and control systematic effects at a level of 0.12 part per billion. Comparison with Penning trap measurements of the electron gyromagnetic anomaly $g_e - 2$ via the Standard Model of particle physics is now limited by the uncertainty in $g_e - 2$; a 2.5σ tension rejects dark photons as the reason for the unexplained part of the muon's magnetic moment at a 99% confidence level. Implications for dark-sector candidates and electron substructure may be a sign of physics beyond the Standard Model that warrants further investigation.

The fine-structure constant α characterizes the strength of the electromagnetic interaction between elementary charged particles. It has been measured by various methods

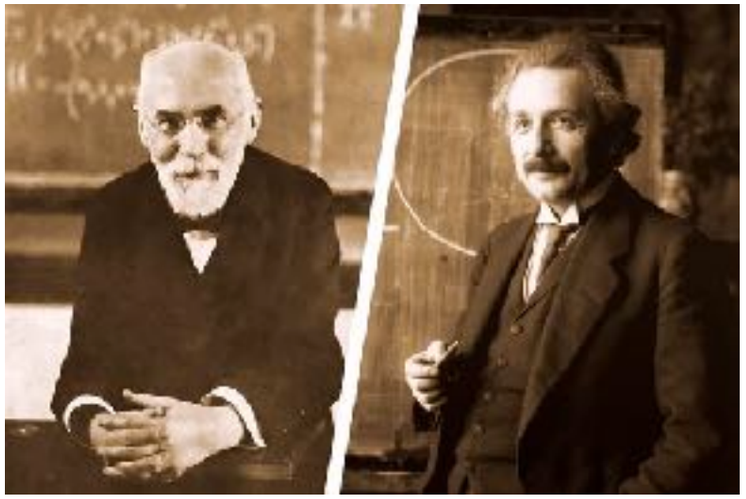
dronic physics (*1*–*3*). This path leads to an accuracy of 0.24 part per billion (ppb) (*4*–*6*) and was until now the most accurate measurement of α.

The Language of Nature

Fast →

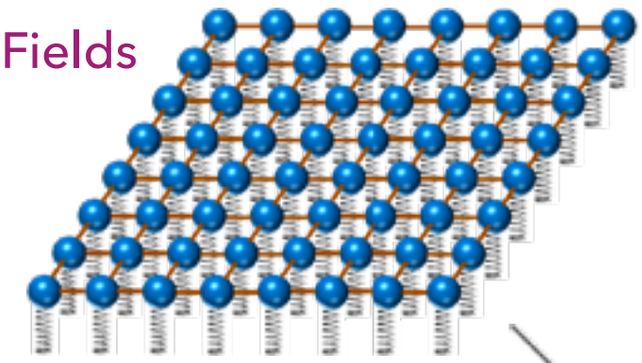
Small ↓

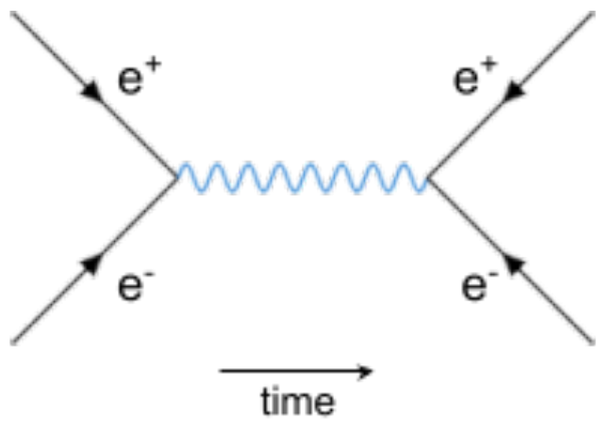
Classical Mechanics	Special Relativity
Quantum Mechanics	**Quantum Field Theory**



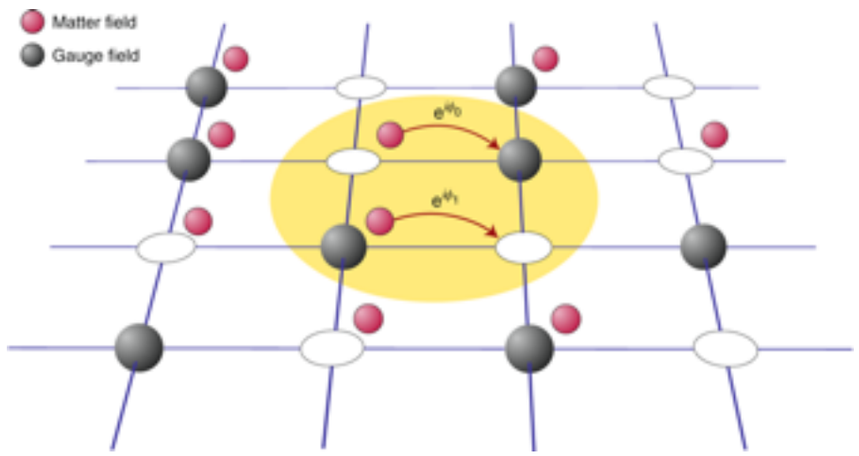
Quantum Field Theory

Fields





A Feynman diagram

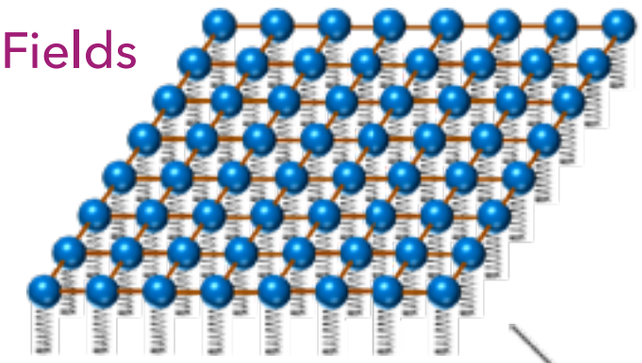


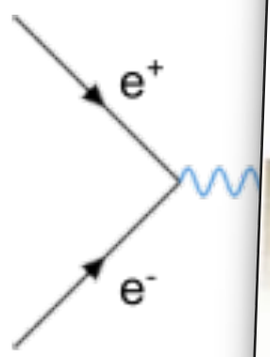
Gauge theory

Emmy Noether

Quantum Field Theory

Fields





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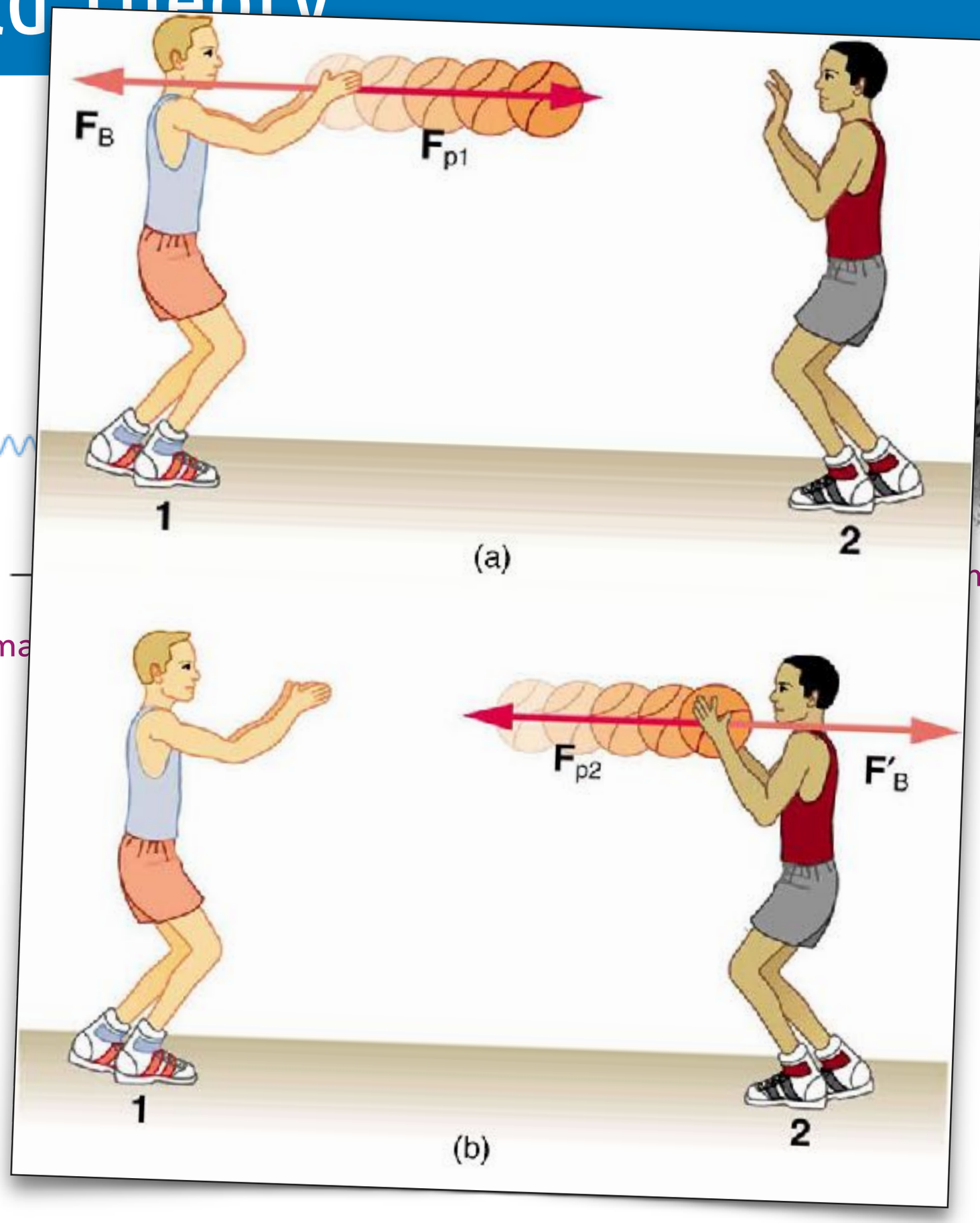
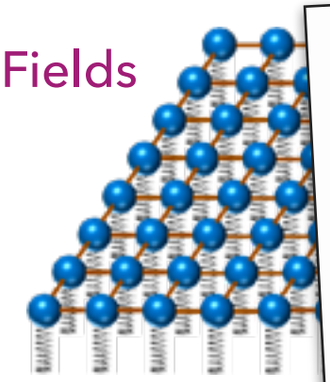
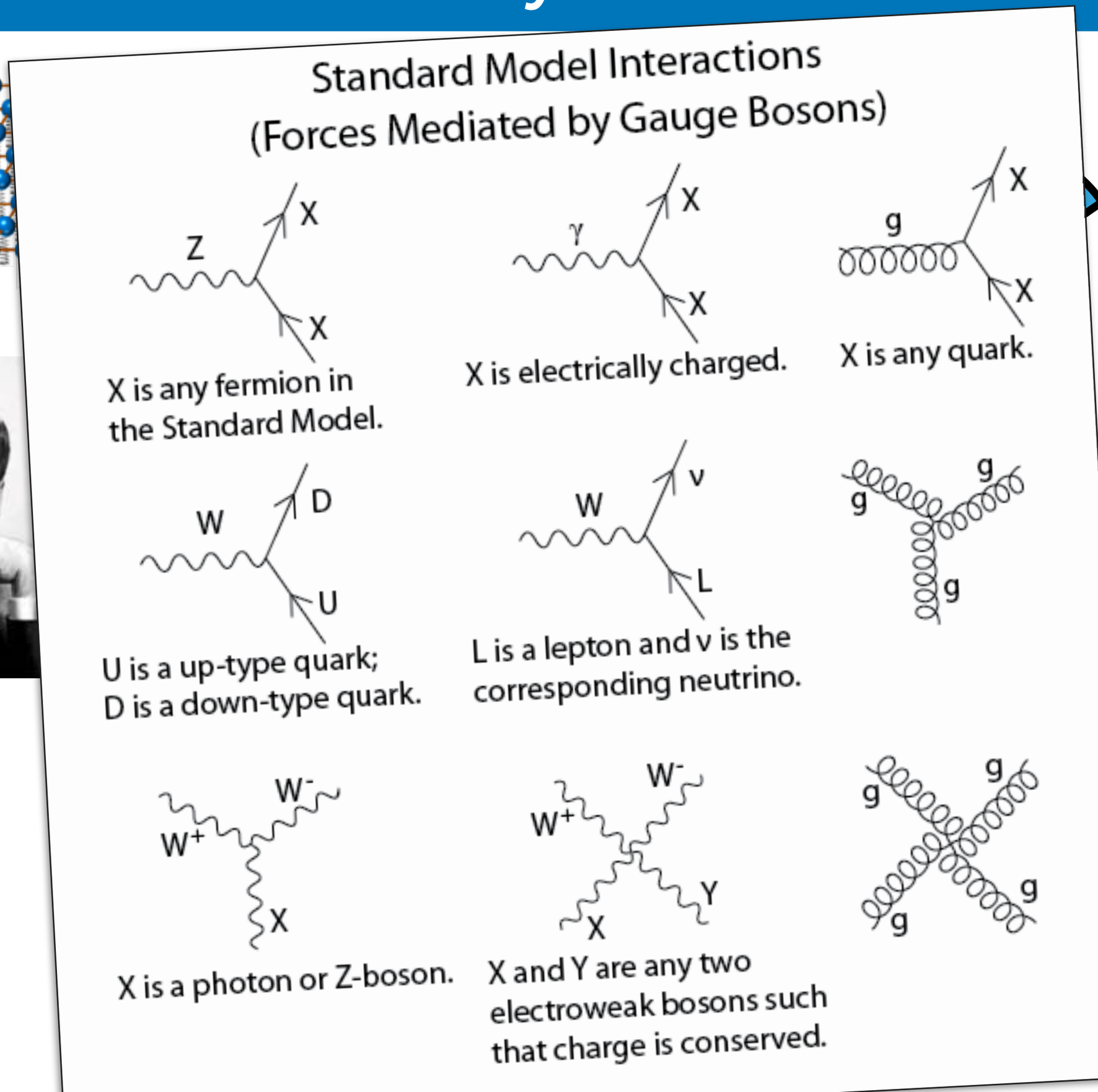


mmy Noether

Quantum Field Theory

Fields

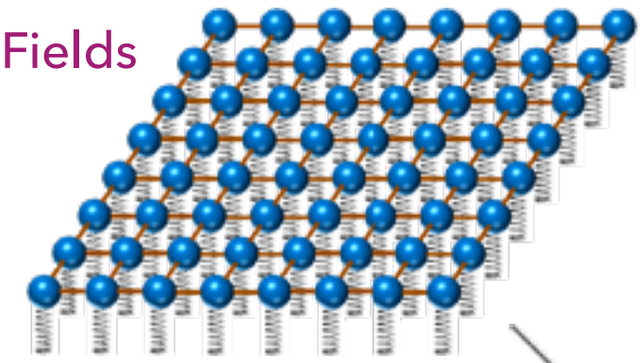


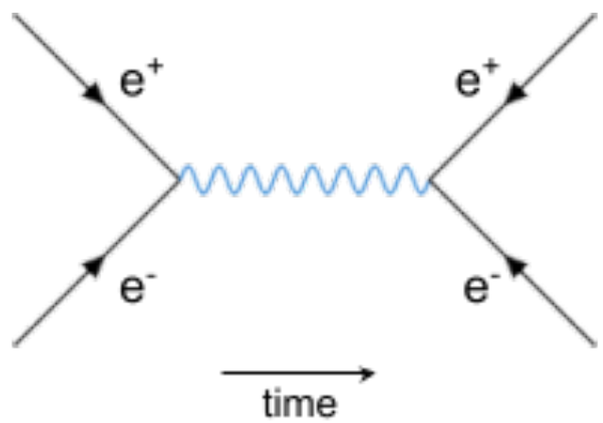


Emmy Noether

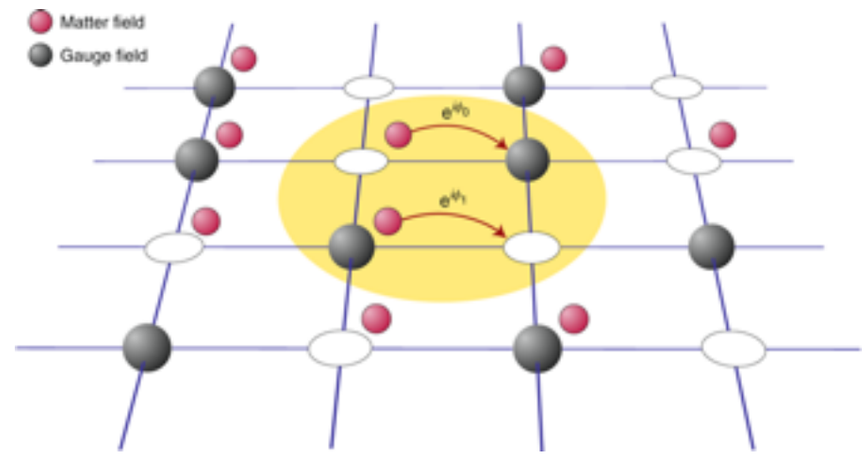
Quantum Field Theory

Fields





A Feynman diagram



Gauge theory

Emmy Noether

A little problem with infinities

The Standard Model

Standard Model of

FUNDAMENTAL PARTICLES AND INTERACTIONS

The Standard Model summarizes the current knowledge in Particle Physics. It is the quantum theory that includes the theory of strong interactions (quantum chromodynamics or QCD) and the unified theory of weak and electromagnetic interactions (electroweak). Gravity is included on this chart because it is one of the fundamental interactions even though not part of the "Standard Model."

FERMIONS

matter constituents
spin = 1/2, 3/2, 5/2, ...

Leptons spin = 1/2

Flavor	Mass GeV/c^2	Electric charge
ν_e electron neutrino	$<1\times10^{-8}$	0
e electron	0.000511	–1
ν_μ muon neutrino	<0.0002	0
μ muon	0.106	–1
ν_τ tau neutrino	<0.02	0
τ tau	1.7771	–1

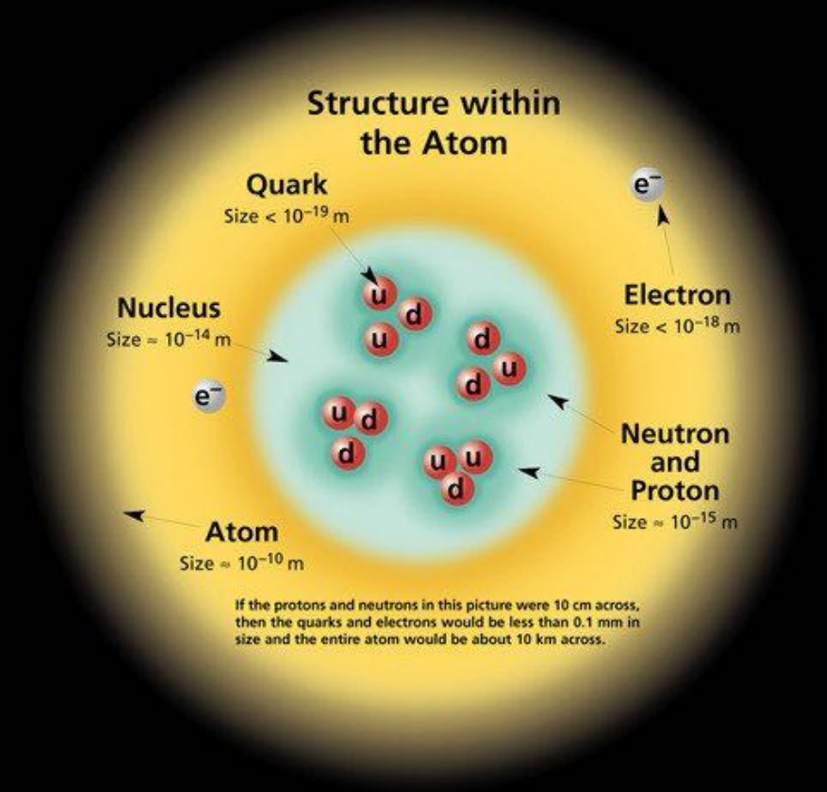
Quarks spin = 1/2

Flavor	Approx. Mass GeV/c^2	Electric charge
u up	0.003	2/3
d down	0.006	–1/3
c charm	1.3	2/3
s strange	0.1	–1/3
t top	175	2/3
b bottom	4.3	–1/3

Spin is the intrinsic angular momentum of particles. Spin is given in units of $\hbar$, which is the quantum unit of angular momentum, where $\hbar = h/2\pi = 6.58\times10^{-25}$ GeV s $= 1.05\times10^{-34}$ J s.

Electric charges are given in units of the proton's charge. In SI units the electric charge of the proton is 1.60×10^{-19} coulombs.

The **energy** unit of particle physics is the electronvolt (eV), the energy gained by one electron in crossing a potential difference of one volt. **Masses** are given in GeV/c^2 (remember $E = mc^2$), where 1 GeV = 10^9 eV = 1.60×10^{-10} joule. The mass of the proton is 0.938 GeV/c^2 = 1.67×10^{-27} kg.



BOSONS

force carriers
spin = 0, 1, 2, ...

Unified Electroweak spin = 1

Name	Mass GeV/c^2	Electric charge
γ photon	0	0
W^-	80.4	–1
W^+	80.4	+1
Z^0	91.187	0

Strong (color) spin = 1

Name	Mass GeV/c^2	Electric charge
g gluon	0	0

Color Charge
Each quark carries one of three types of "strong charge," also called "color charge." These charges have nothing to do with the colors of visible light. There are eight possible types of color charge for gluons. Just as electrically-charged particles interact by exchanging photons, in strong interactions color-charged particles interact by exchanging gluons. Leptons, photons, and ***W*** and ***Z*** bosons have no strong interactions and hence no color charge.

Quarks Confined in Mesons and Baryons
One cannot isolate quarks and gluons; they are confined in color-neutral particles called **hadrons**. This confinement (binding) results from multiple exchanges of gluons among the color-charged constituents. As color-charged particles (quarks and gluons) move apart, the energy in the color-force field between them increases. This energy eventually is converted into additional quark-antiquark pairs (see figure below). The quarks and antiquarks then combine into hadrons; these are the particles seen to emerge. Two types of hadrons have been observed in nature: **mesons** $q\bar{q}$ and **baryons** qqq.

Residual Strong Interaction
The strong binding of color-neutral protons and neutrons to form nuclei is due to residual strong interactions between their color-charged constituents. It is similar to the residual electrical interaction that binds electrically neutral atoms to form molecules. It can also be viewed as the exchange of mesons between the hadrons.

PROPERTIES OF THE INTERACTIONS

Property \ Interaction	Gravitational	Weak (Electroweak)	Electromagnetic (Electroweak)	Strong: Fundamental	Strong: Residual
Acts on:	Mass – Energy	Flavor	Electric Charge	Color Charge	See Residual Strong Interaction Note
Particles experiencing:	All	Quarks, Leptons	Electrically charged	Quarks, Gluons	Hadrons
Particles mediating:	Graviton (not yet observed)	W^+ W^- Z^0	γ	Gluons	Mesons
Strength relative to electromag for two u quarks at: 10^{-18} m	10^{-41}	0.8	1	25	Not applicable to quarks
Strength relative to electromag for two u quarks at: 3×10^{-17} m	10^{-41}	10^{-4}	1	60	
for two protons in nucleus	10^{-36}	10^{-7}	1	Not applicable to hadrons	20

Baryons qqq and Antibaryons $\bar{q}\bar{q}\bar{q}$
Baryons are fermionic hadrons.
There are about 120 types of baryons.

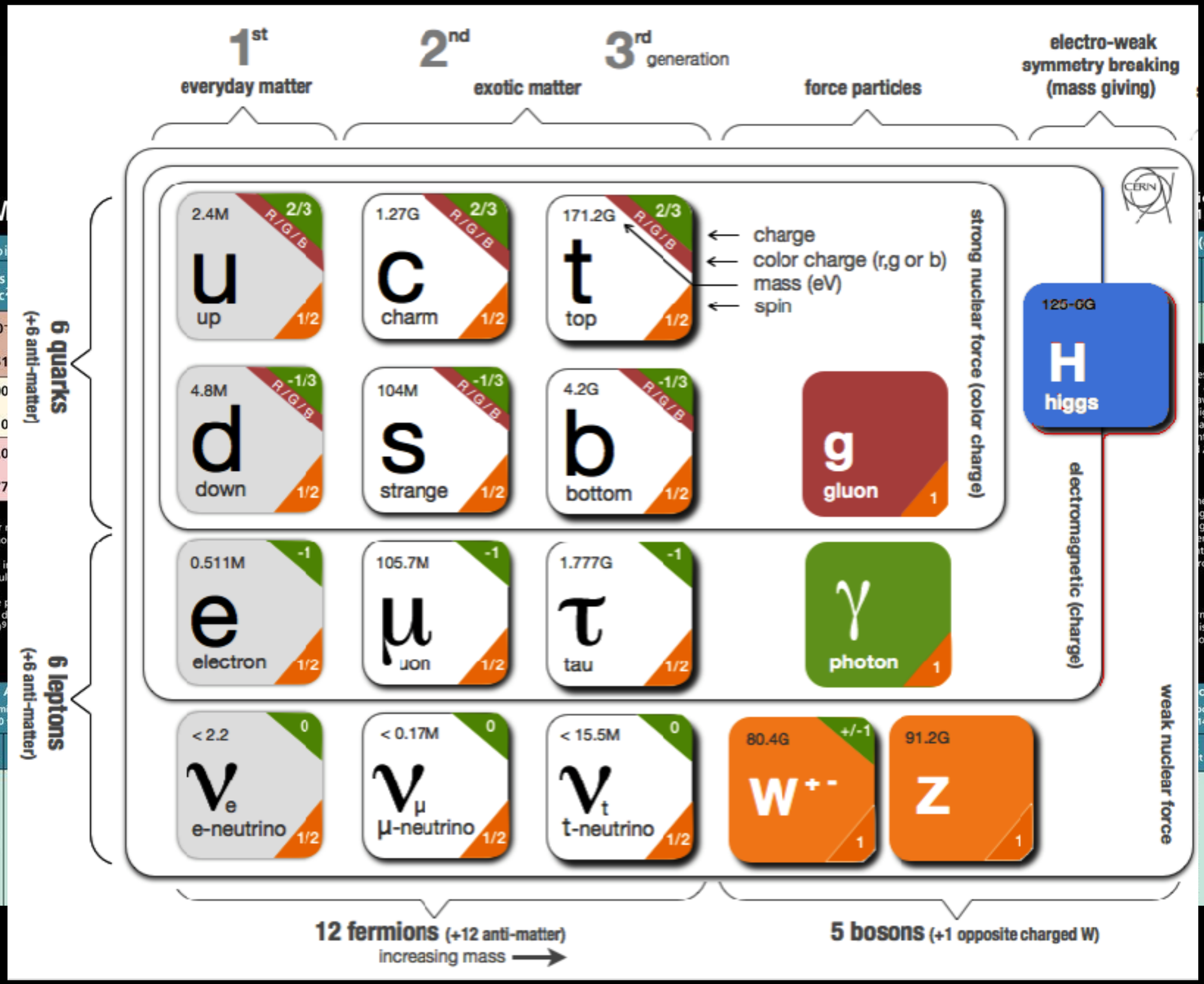
Symbol	Name	Quark content	Electric charge	Mass GeV/c^2	Spin
p	proton	uud	1	0.938	1/2
$\bar{p}$	anti-proton	$\bar{u}\bar{u}\bar{d}$	–1	0.938	1/2
n	neutron	udd	0	0.940	1/2
Λ	lambda	uds	0	1.116	1/2
Ω^-	omega	sss	–1	1.672	3/2

Mesons $q\bar{q}$
Mesons are bosonic hadrons.
There are about 140 types of mesons.

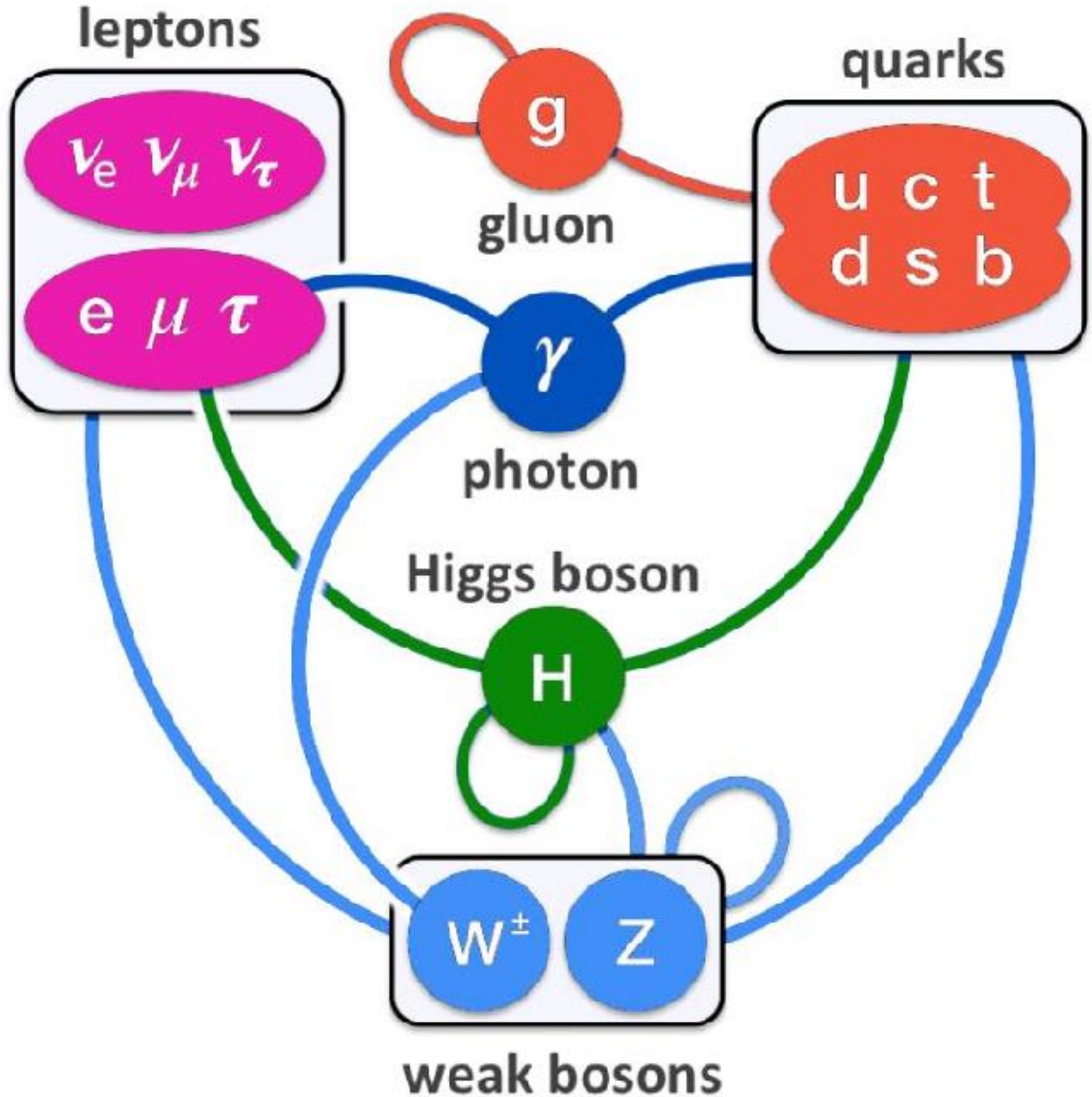
Symbol	Name	Quark content	Electric charge	Mass GeV/c^2	Spin
π^+	pion	$u\bar{d}$	+1	0.140	0
K^-	kaon	$s\bar{u}$	–1	0.494	0
ρ^+	rho	$u\bar{d}$	+1	0.770	1
B^0	B-zero	$d\bar{b}$	0	5.279	0
η_c	eta-c	$c\bar{c}$	0	2 .980	0

The Missing Piece

The Standard Model



The Standard Model



FERMIO

Leptons spin =		
Flavor	Mass GeV/c²	El ch
ν_e electron neutrino	<1×10⁻⁸	
e electron	0.000511	
ν_μ muon neutrino	<0.0002	
μ muon	0.106	
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τ tau	1.7771	

Spin is the intrinsic angular momen
quantum unit of angular momentu

Electric charges are given in units
the proton is 1.60×10^{-19} coulombs.

The **energy** unit of particle physics
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$E = mc^2$), where 1 GeV = 10^9 eV = 1
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There are about 120 types o			
Symbol	Name	Quark content	Electri charge
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$\bar{p}$	anti-proton	$\bar{u}\bar{u}\bar{d}$	–1
n	neutron	udd	0
Λ	lambda	uds	0
Ω^-	omega	sss	–1

S

force carriers
spin = 0, 1, 2, ...

Strong (color) spin = 1		
Name	Mass GeV/c²	Electric charge
g gluon	0	0

Color Charge
Each quark carries one of three types of
"strong charge," also called "color charge."
These charges have nothing to do with the
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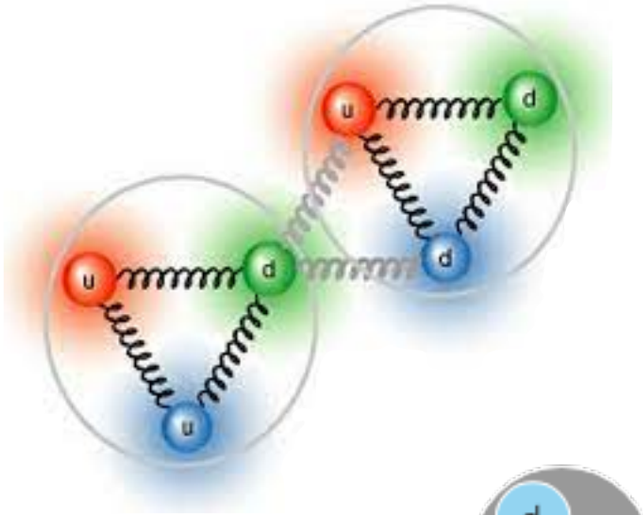
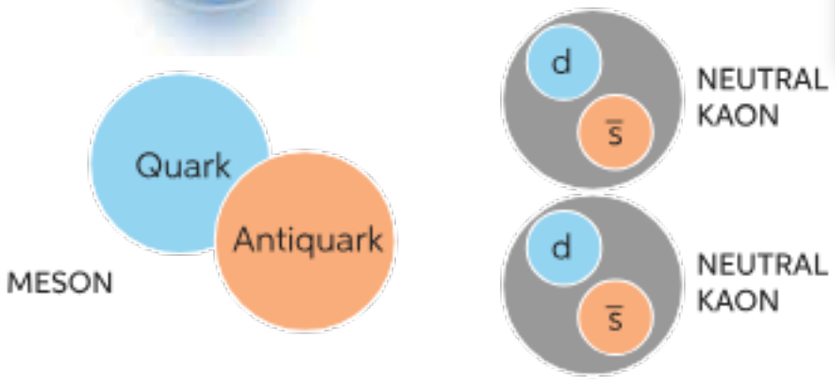
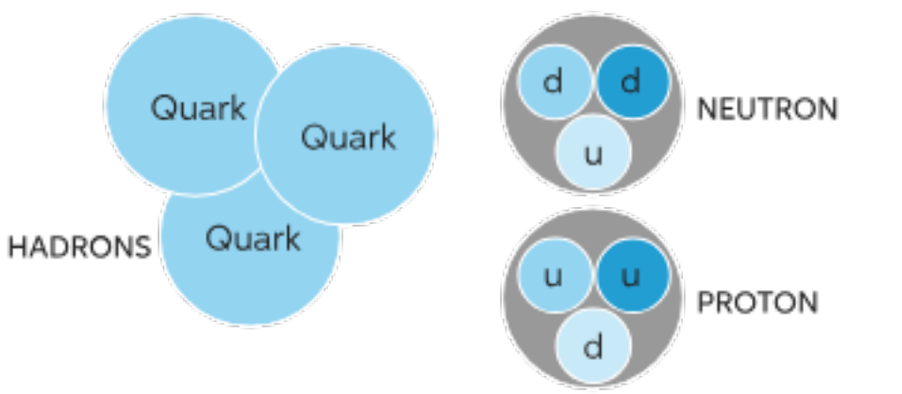
constituents. It is similar to the residual elec-
oms to form molecules. It can also be
hadrons.

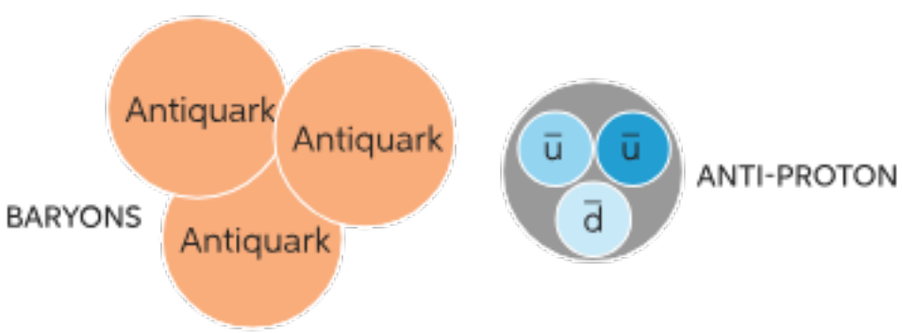
Mesons $q\bar{q}$				
Mesons are bosonic hadrons.				
There are about 140 types of mesons.				
Name	Quark content	Electric charge	Mass GeV/c²	Spin
pion	$u\bar{d}$	+1	0.140	0
kaon	$s\bar{u}$	–1	0.494	0
rho	$u\bar{d}$	+1	0.770	1
B-zero	$d\bar{b}$	0	5.279	0
eta-c	$c\bar{c}$	0	2 .980	0

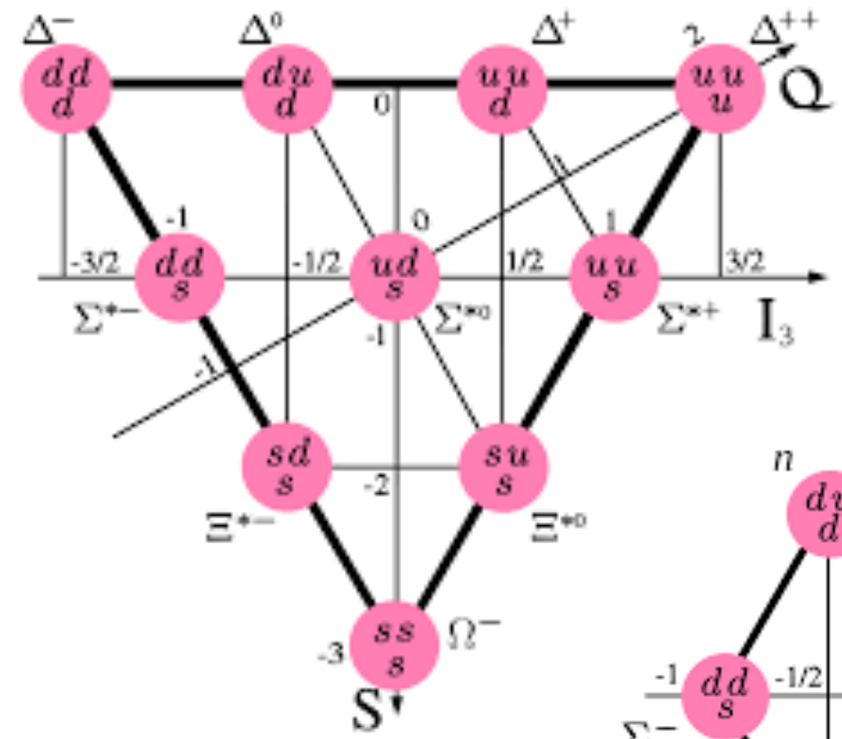
Bound states

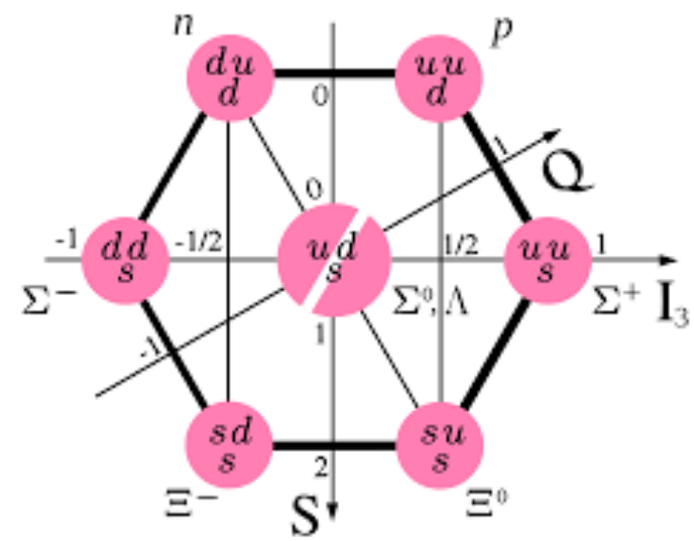




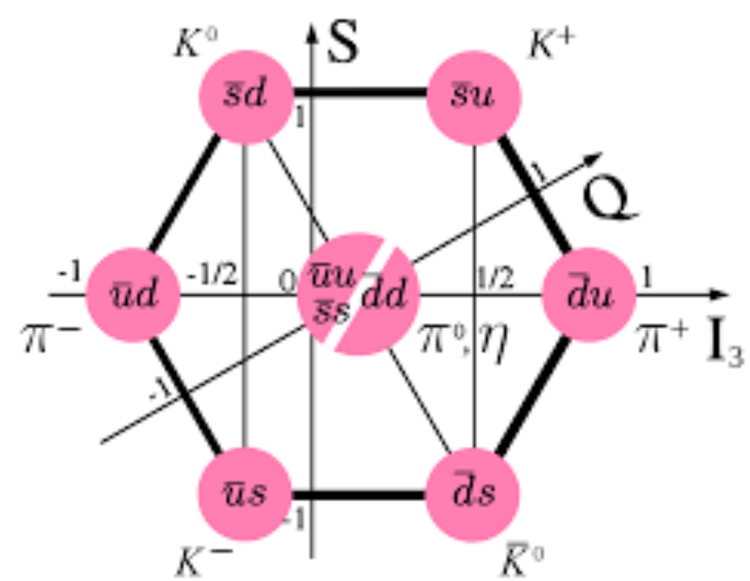








Baryons



Mesons

How much do we know?

Then

Every object of creation is made of atoms (paramāṇu) which in turn connect with each other to form molecules (aṇu). Atoms are eternal, and their combinations constitute the empirical material world.

Now

Particle Data Group https://pdg.lbl.gov/

A good set of units

SI units are NOT useful.
Too big!

Make life simple by setting

$$c = \hbar = 1$$
$$1\ eV = 1.6 \times 10^{-19} J$$

Actually, particle theorists are lazy (or clever, depending on your point of view)—they seldom include the c's and $\hbar$'s ($\hbar \equiv h/2\pi$) in their formulas. You're just supposed to fit them in for yourself at the end, to make the dimensions come out right. As they say in the business, "set $c = \hbar = 1$." This amounts to working in units such that time is measured in centimeters and mass and energy in inverse centimeters; the unit of time is the time it takes light to travel 1 centimeter, and the unit of energy is the energy of a photon whose wavelength is 2π centimeters.

Conversion Factor	$\hbar = c = 1$ Units	Actual Dimension
$1\ \text{kg} = 5.61 \times 10^{26}\ \text{GeV}$	GeV	$\frac{\text{GeV}}{c^2}$
$1\ \text{m} = 5.07 \times 10^{15}\ \text{GeV}^{-1}$	GeV^{-1}	$\frac{\hbar c}{\text{GeV}}$
$1\ \text{sec} = 1.52 \times 10^{24}\ \text{GeV}^{-1}$	GeV^{-1}	$\frac{\hbar}{\text{GeV}}$
$e = \sqrt{4\pi\alpha}$	—	$(\hbar c)^{1/2}$

By choosing units with $\hbar = c = 1$, it becomes unnecessary to write $\hbar$ and c explicitly in the formulas, thus saving a lot of time and trouble. We can always use dimensional analysis to work out unambiguously where the $\hbar$'s and c's enter any formula. Hence, with a slight but permissible laziness, it is customary to speak of mass (m), momentum (mc), and energy (mc^2) all in terms of GeV, and to measure length ($\hbar/mc$) and time ($\hbar/mc^2$) in units of GeV^{-1}.

$$1\ \text{TeV} = 10^3\ \text{GeV} = 10^6\ \text{MeV} = 10^9\ \text{KeV} = 10^{12}\ \text{eV}$$
$$1\ \text{fermi} \equiv 1\ \text{F} = 10^{-13}\ \text{cm} = 5.07\ \text{GeV}^{-1}$$
$$(1\ \text{F})^2 = 10\ \text{mb} = 10^4\ \mu\text{b} = 10^7\ \text{nb} = 10^{10}\ \text{pb}$$
$$(1\ \text{GeV})^{-2} = 0.389\ \text{mb}$$

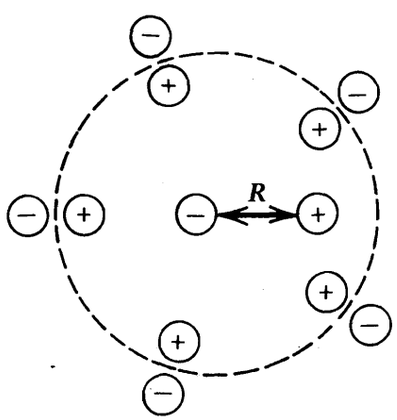
The charge of an electron

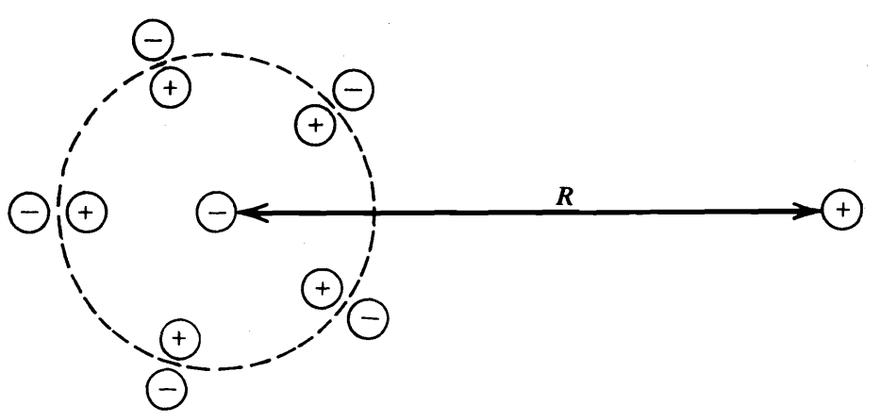
So far, we have not considered the elementary charge e, which measures how strongly electrons, say, interact electromagnetically with each other. To obtain a dimensionless measure of the strength of this interaction, we compare the electrostatic energy of repulsion between two electrons one natural unit of length apart with the rest mass energy of an electron:

$$\boxed{\alpha = \frac{1}{4\pi}\frac{e^2}{(\hbar/mc)}\Big/mc^2 = \frac{e^2}{4\pi\hbar c} \approx \frac{1}{137}.}$$

we have adopted the rationalized Heaviside–Lorentz system of electromagnetic units. That is, the 4π factors appear in the force equations rather than in the Maxwell equations, and ε_0 is set equal to unity. This choice, which is conventional in particle physics, reduces Maxwell's equations to their simplest possible form. The value of α is, of course, the same in all systems of units, but the numerical value of e is different.

For historical reasons, α is known as the fine structure constant. Unfortunately, this name conveys a false impression. We have seen that the charge of an electron is not strictly constant but varies with distance because of quantum effects; hence α must be regarded as a variable, too. The value $\frac{1}{137}$ is the asymptotic value of α





β decay

$$A \rightarrow B + e^-$$

Conservation of charge requires that B carry one more unit of positive charge than A. (We now realize that the underlying process here is the conversion of a neutron, in A, into a proton, in B; but remember that in 1930 the neutron had not yet been discovered.) Thus the 'daughter' nucleus (B) lies one position farther along on the periodic table. There are many examples of beta decay: potassium goes to calcium ($^{40}_{19}\mathrm{K} \rightarrow ^{40}_{20}\mathrm{Ca}$), copper goes to zinc ($^{64}_{29}\mathrm{Cu} \rightarrow ^{64}_{30}\mathrm{Zn}$), tritium goes to helium ($^{3}_{1}\mathrm{H} \rightarrow ^{3}_{2}\mathrm{He}$), and so on.*

Now, it is a characteristic of two-body decays ($A \rightarrow B + C$) that the outgoing energies are kinematically determined, in the center-of-mass frame. Specifically, if the 'parent' nucleus (A) is at rest, so that B and e come out back-to-back with equal and opposite momenta, then conservation of energy dictates that the electron energy is

$$E = \left(\frac{m_A^2 - m_B^2 + m_e^2}{2m_A} \right) c^2$$

In modern terminology, the fundamental beta decay process isess is

$$n \rightarrow p^+ + e^- + \overline{\nu}$$

(neutron goes to proton plus electron plus antineutrino).

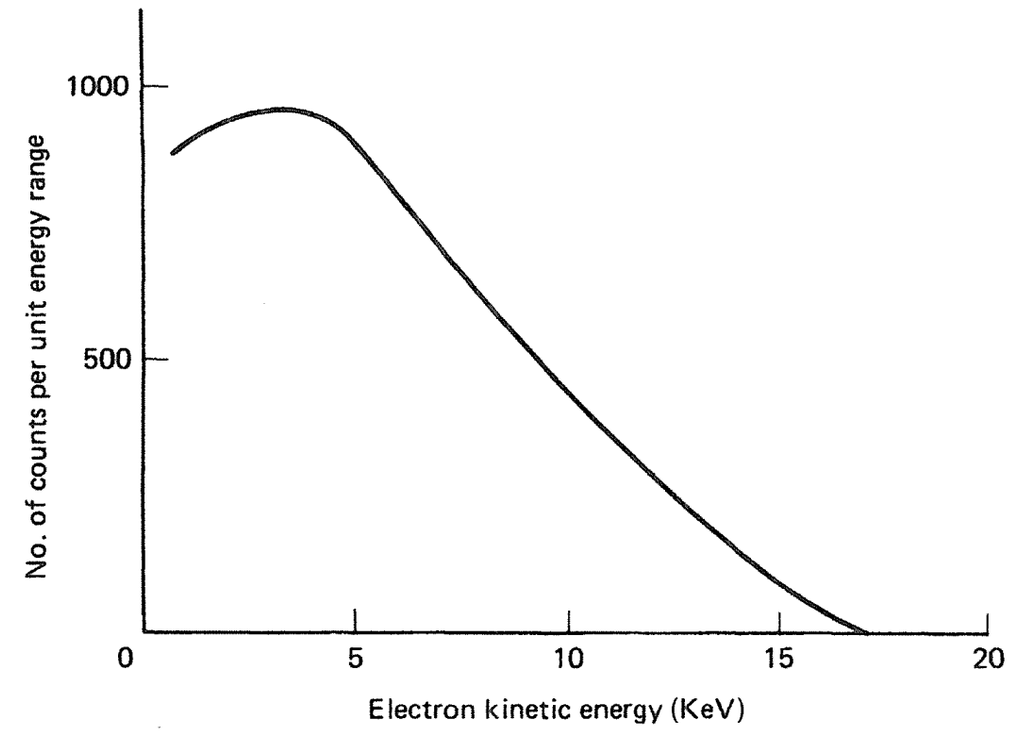
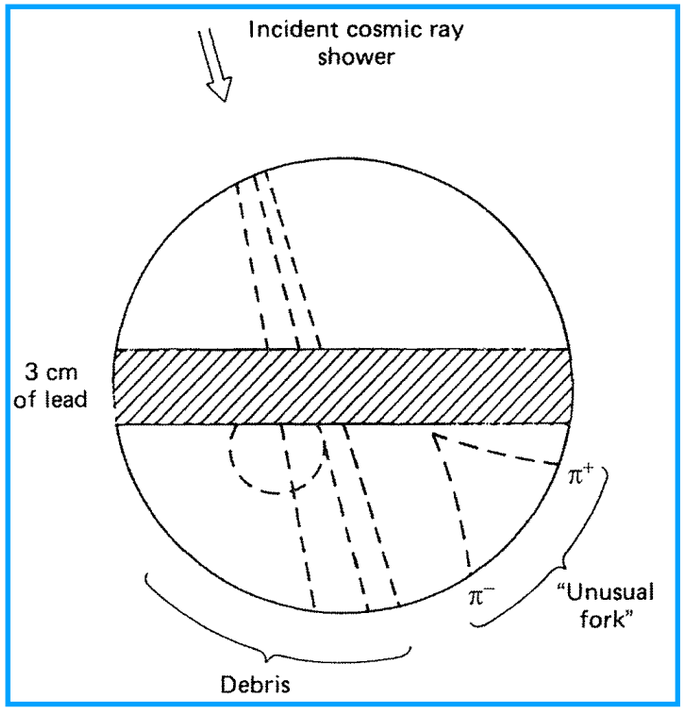


Table 1.1 The lepton family, 1962–1976

	Lepton number	Electron number	Muon number
Leptons			
e^-	1	1	0
ν_e	1	1	0
μ^-	1	0	1
ν_μ	1	0	1
Antileptons			
e^+	−1	−1	0
$\overline{\nu}_e$	−1	−1	0
μ^+	−1	0	−1
$\overline{\nu}_\mu$	−1	0	−1

Strange Particles

For a brief period in 1947, it was possible to believe that the major problems of elementary particle physics were solved.



In December 1947, Rochester and Butler discovered a neutral particle decaying to two pions

$$K^0 \rightarrow \pi^+ + \pi^-$$

In 1949, another Keon was discovered

$$K^+ \rightarrow \pi^+ + \pi^+ + \pi^-$$

In due course, many more mesons were discovered – the η, the ϕ, the ω, the ρ's, and so on.

In 1950, another Baryon was discovered

$$\Lambda \rightarrow p^+ + \pi^-$$

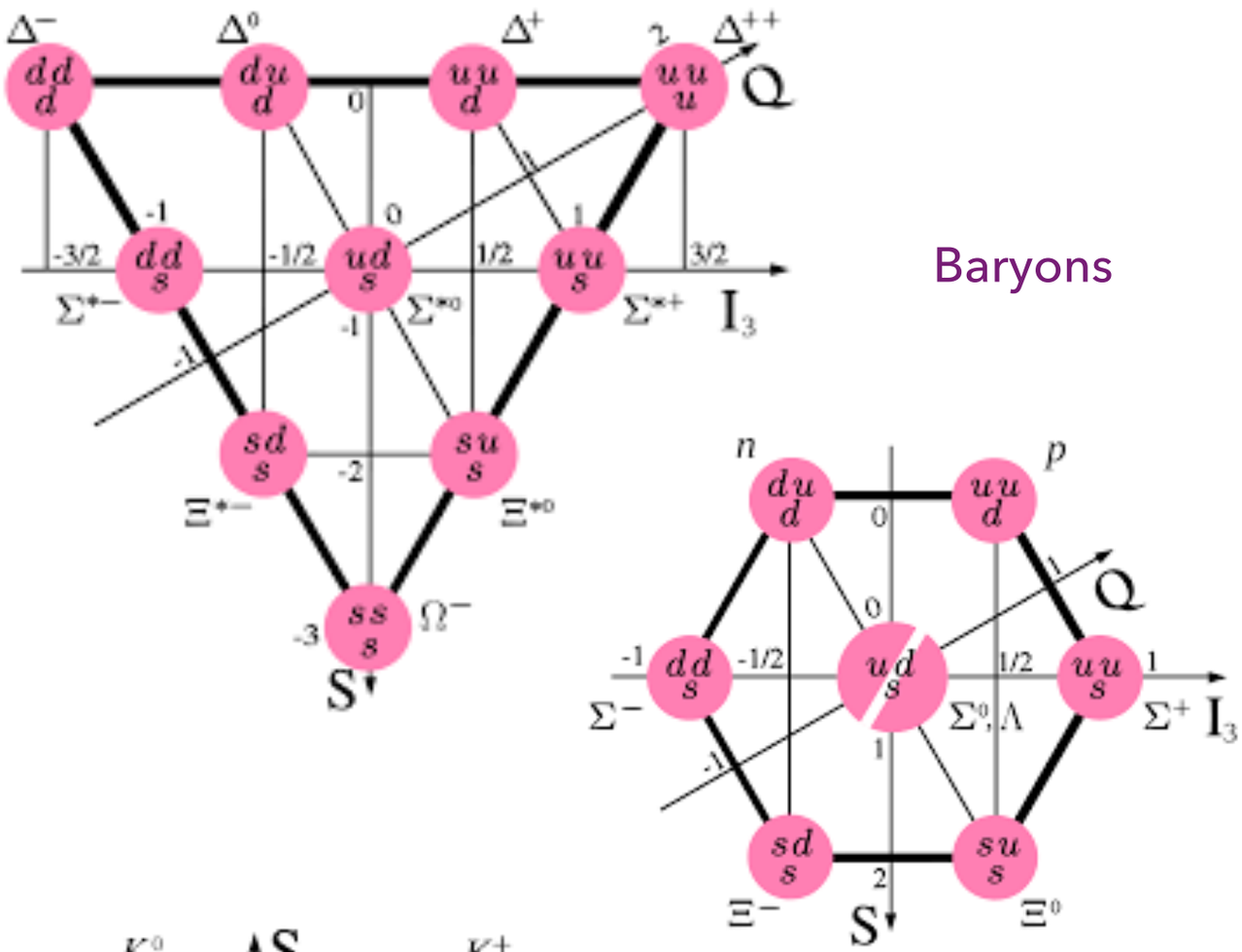
Not only were the new particles unexpected; there is a more technical sense in which they seemed 'strange': they are *produced* copiously (on a time scale of about 10^{-23} seconds), but they *decay* relatively slowly (typically about 10^{-10} seconds). This suggested that the mechanism involved in their production is entirely different from that which governs their disintegration. In modern language, the strange particles are *produced* by the *strong* force (the same one that holds the nucleus together), but they *decay* by the *weak* force (the one that accounts for beta decay and all other neutrino processes).

'Why is the proton stable?' Why, for example, doesn't it decay into a positron and a photon:

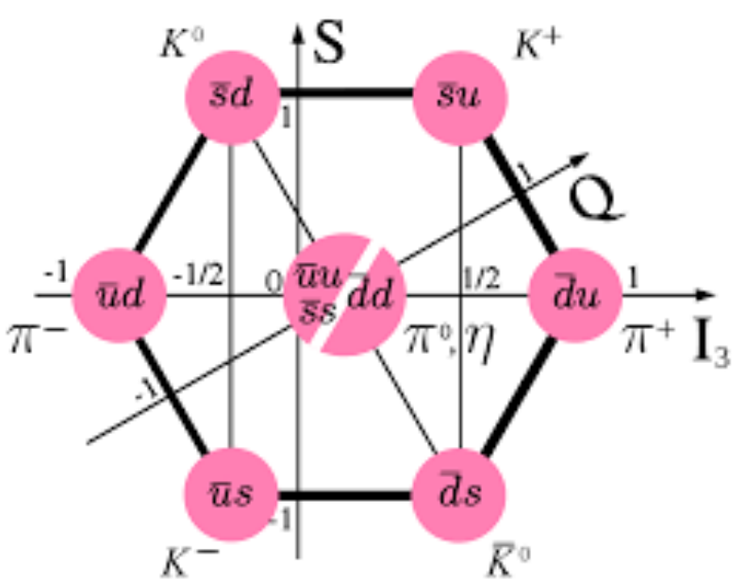
$$p^+ \rightarrow e^+ + \gamma$$

Baryon Number

There are patterns

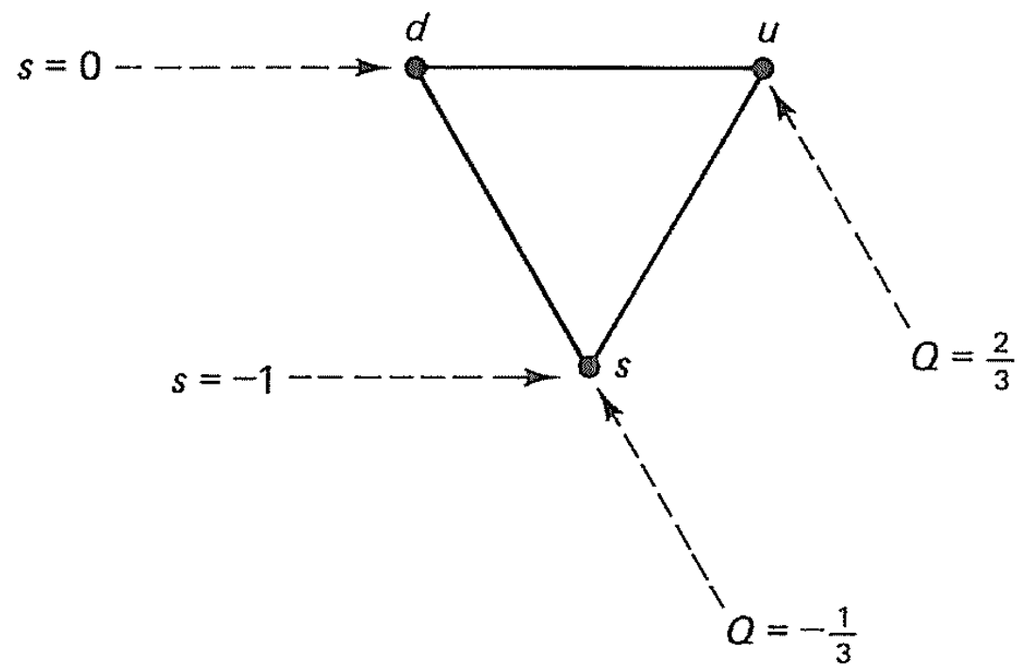


Baryons



Mesons

The Quark Model

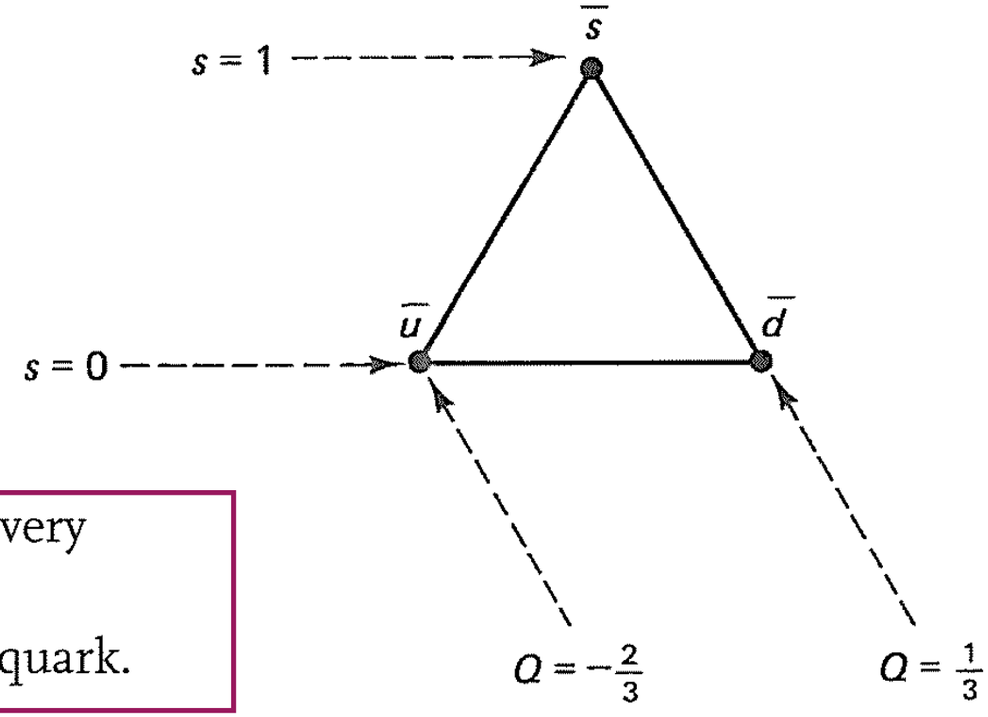


The quarks

$$p = uud$$

$$n = udd$$

$$\Delta^{++} = uuu.$$

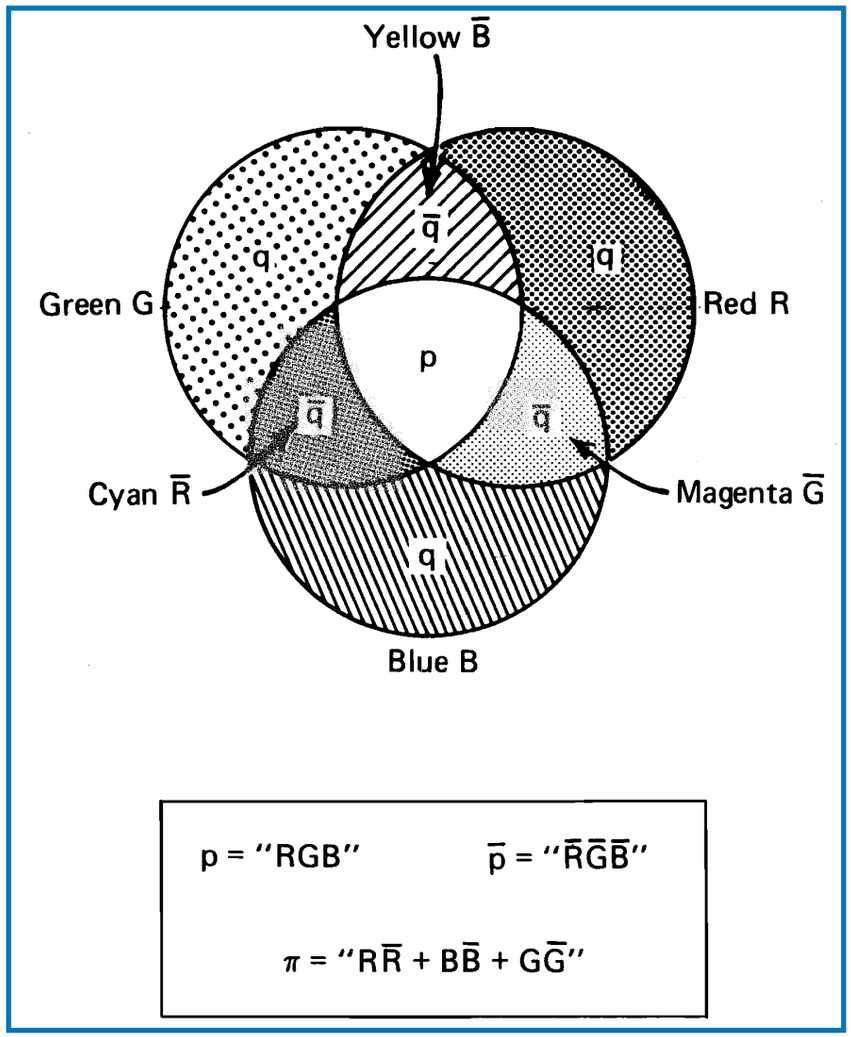


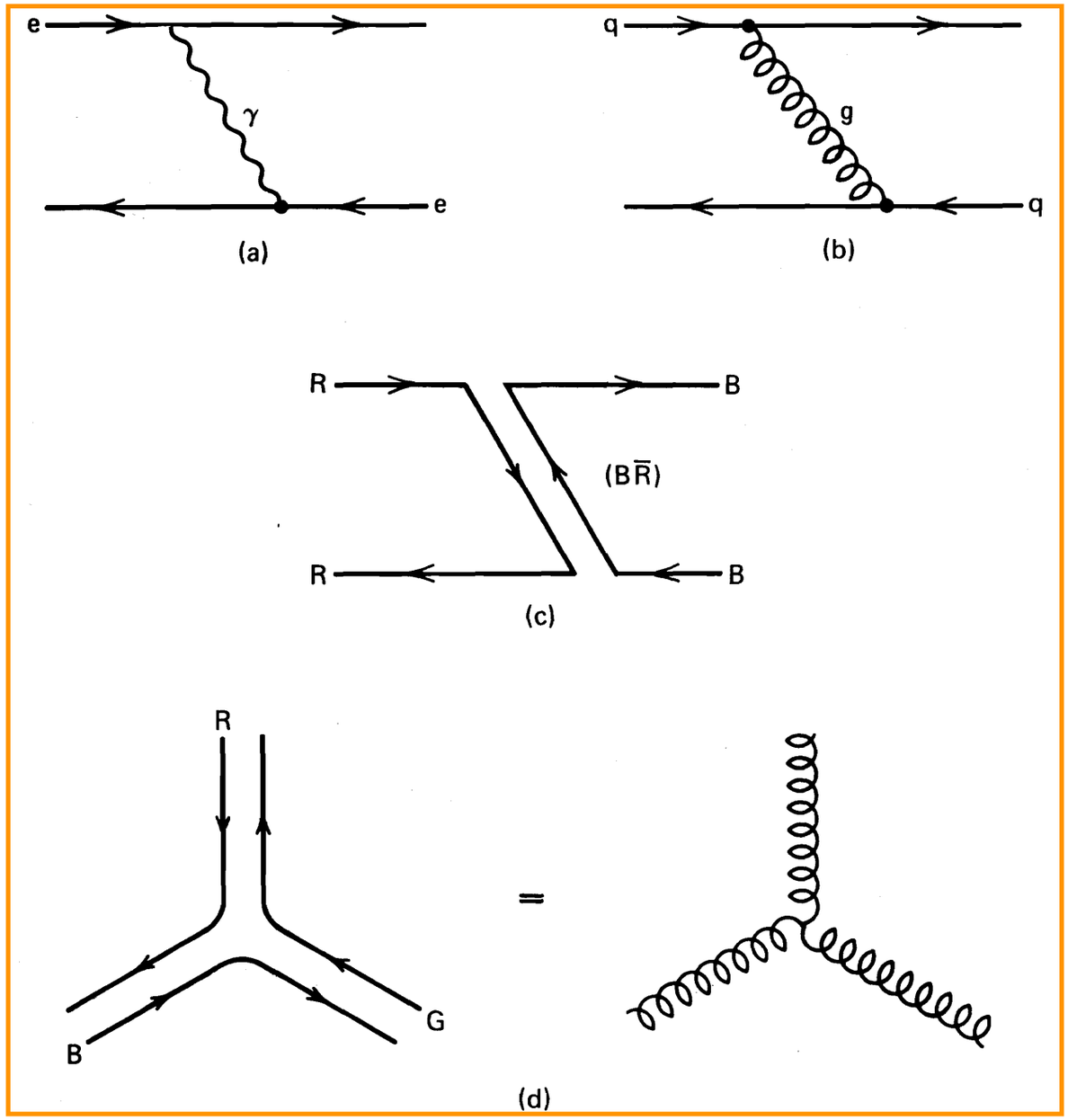
The antiquarks

1. Every baryon is composed of three quarks (and every *anti*baryon is composed of three *anti*quarks).
2. Every meson is composed of a quark and an antiquark.

The Quark Model

The quarks have colour
but the "world" is colourless





1. Every baryon is composed of three quarks (and every *anti*baryon is composed of three *anti*quarks).
2. Every meson is composed of a quark and an antiquark.

So, what do we have?

Charge: All three interactions, of course, conserve electric charge. In the case of the weak interactions, the lepton (or quark) that comes out may not have the same charge as the one that went in, but if so, the difference is carried away by the W.

Color: The electromagnetic and weak interactions do not affect color. At a strong vertex the quark color does change, but the difference is carried off by the gluon. (The direct gluon–gluon couplings also conserve color.) However, since naturally occurring particles are always colorless, the observable manifestation of color conservation is pretty trivial: zero in, zero out.

Baryon number: In all the primitive vertices, if a quark goes *in*, a quark comes *out*, so the total number of quarks present is a constant. In this arithmetic we count *anti*quarks as *negative*, so that, for example, at the vertex $q + \bar{q} \to g$ the quark number is zero before and zero after. Of course, we never see individual quarks, only baryons (with quark number 3), antibaryons (quark number -3), and mesons (quark number zero). So, in practice, it is more convenient to speak of the conservation of baryon number (1 for baryons, -1 for antibaryons, and 0 for everything else). The baryon number is just $\frac{1}{3}$ the quark number. Notice that there is no analogous conservation of *meson number*; since mesons carry zero quark number, a given collision or decay can produce as many mesons as it likes, consistent with conservation of energy.

Lepton number: The strong forces do not touch leptons at all; in an electromagnetic interaction the same particle comes out (accompanied by a photon) as went in; and in the weak interactions if a lepton goes in, a lepton comes out (not necessarily the *same* one, this time). So, lepton number is absolutely conserved. Until recently there appeared to be no cross-generation mixing among the leptons, so electron number, muon number, and tau number were all separately conserved. This remains true in most cases, but neutrino oscillations indicate that it is not absolute.*

So, what do we have?

Flavor: What about quark *flavor?* Flavor is conserved at a strong or electromagnetic vertex, but *not* at a weak vertex, where an up-quark may turn into a down quark or a strange quark, with nothing at all picking up the lost upness or supplying the 'gained' downness or strangeness. Because the weak forces *are* so weak, we say that the various flavors are *approximately* conserved. In fact, as you may remember, it was precisely this approximate conservation that led Gell-Mann to introduce the notion of strangeness in the first place. He 'explained' the fact that strange particles are always produced in pairs:

$$\pi^-(d\bar{u}) + p^+(uud) \rightarrow K^+(u\bar{s}) + \Sigma^-(dds)$$

for instance, but

$$\pi^-(d\bar{u}) + p^+(uud) \not\rightarrow \pi^+(u\bar{d}) + \Sigma^-(dds)$$

by arguing that the latter violates conservation of strangeness. (Actually, this *is* a possible *weak* interaction, but it will never be seen in the laboratory, because it must compete against enormously more probable strong processes that do conserve strangeness.) In *decays*, however, the nonconservation of strangeness is very conspicuous, because for many particles this is the only way they *can* decay; there *is* no competition from strong or electromagnetic processes. The Λ, for instance, is the lightest strange baryon; if it is to decay, it must go to n (or p) plus something. But the lightest strange meson is the K, and n (or p) plus K weighs substantially more than the Λ. If the Λ decays at all (and it *does* as we know: $\Lambda \rightarrow p^+ + \pi^-$ 64% of the time, and $\Lambda \rightarrow n + \pi^0$ 36% of the time), then strangeness cannot be conserved, and the reaction must proceed by the weak interaction. By contrast, the Δ^0 (with a strangeness of zero) can go to $p^+ + \pi^-$ or $n + \pi^0$ by the *strong* interaction, and its lifetime is accordingly much shorter.

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Which are allowed?

Home work

(a) $p + \bar{p} \rightarrow \pi^+ + \pi^0$

(b) $\eta \rightarrow \gamma + \gamma$

(c) $\Sigma^0 \rightarrow \Lambda + \pi^0$

(d) $\Sigma^- \rightarrow n + \pi^-$

(e) $e^+ + e^- \rightarrow \mu^+ + \mu^-$

(f) $\mu^- \rightarrow e^- + \bar{\nu}_e$

(g) $\Delta^+ \rightarrow p + \pi^0$

(h) $\bar{\nu}_e + p \rightarrow n + e^+$

(i) $e + p \rightarrow \nu_e + \pi^0$

(j) $p + p \rightarrow \Sigma^+ + n + K^0 + \pi^+ + \pi^0$

(k) $p \rightarrow e^+ + \gamma$

(l) $p + p \rightarrow p + p + p + \bar{p}$

(m) $n + \bar{n} \rightarrow \pi^+ + \pi^- + \pi^0$

(n) $\pi^+ + n \rightarrow \pi^- + p$

(o) $K^- \rightarrow \pi^- + \pi^0$

(p) $\Sigma^+ + n \rightarrow \Sigma^- + p$

(q) $\Sigma^0 \rightarrow \Lambda + \gamma$

(r) $\Xi^- \rightarrow \Lambda + \pi^-$

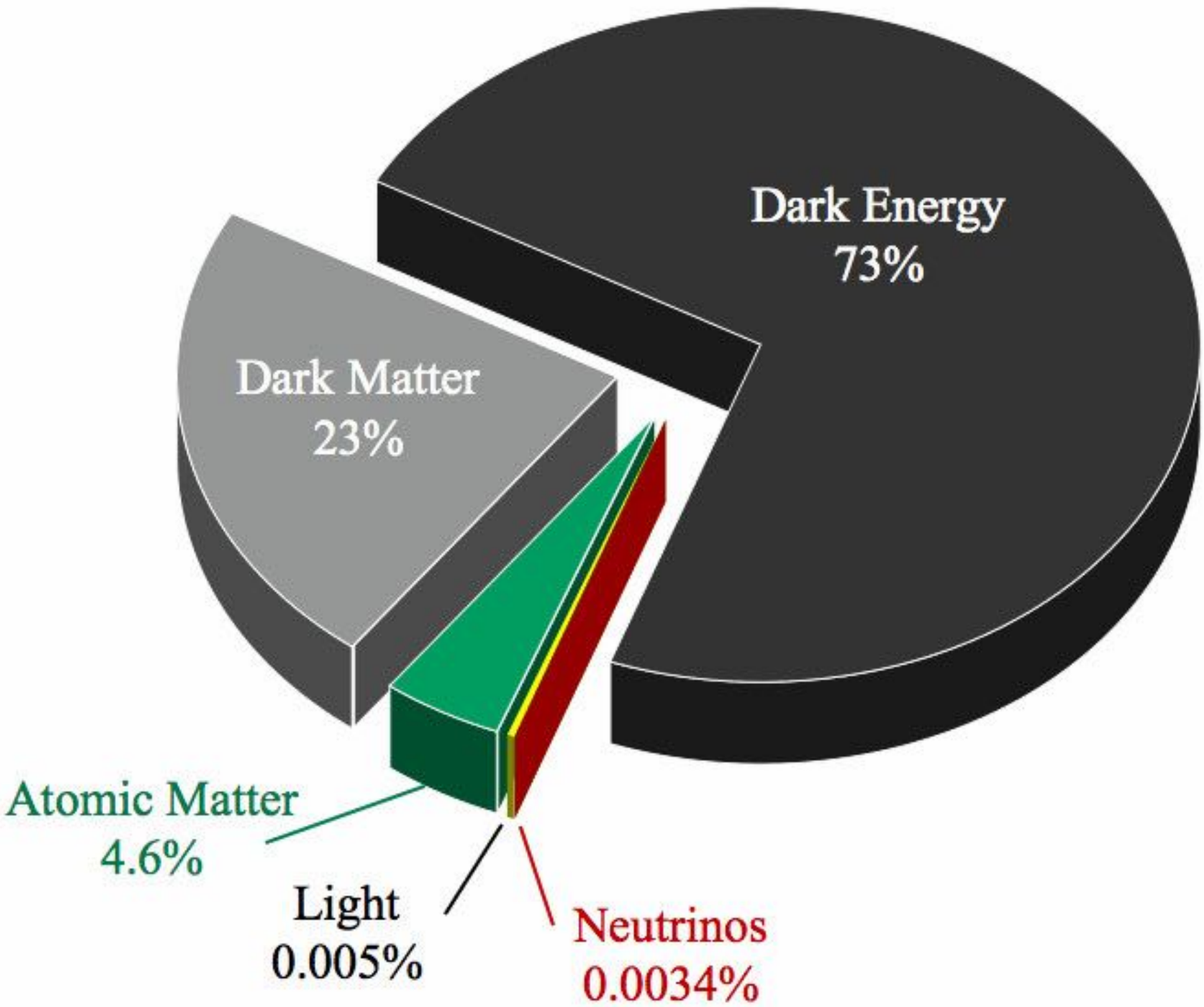
(s) $\Xi^0 \rightarrow p + \pi^-$

(t) $\pi^- + p \rightarrow \Lambda + K^0$

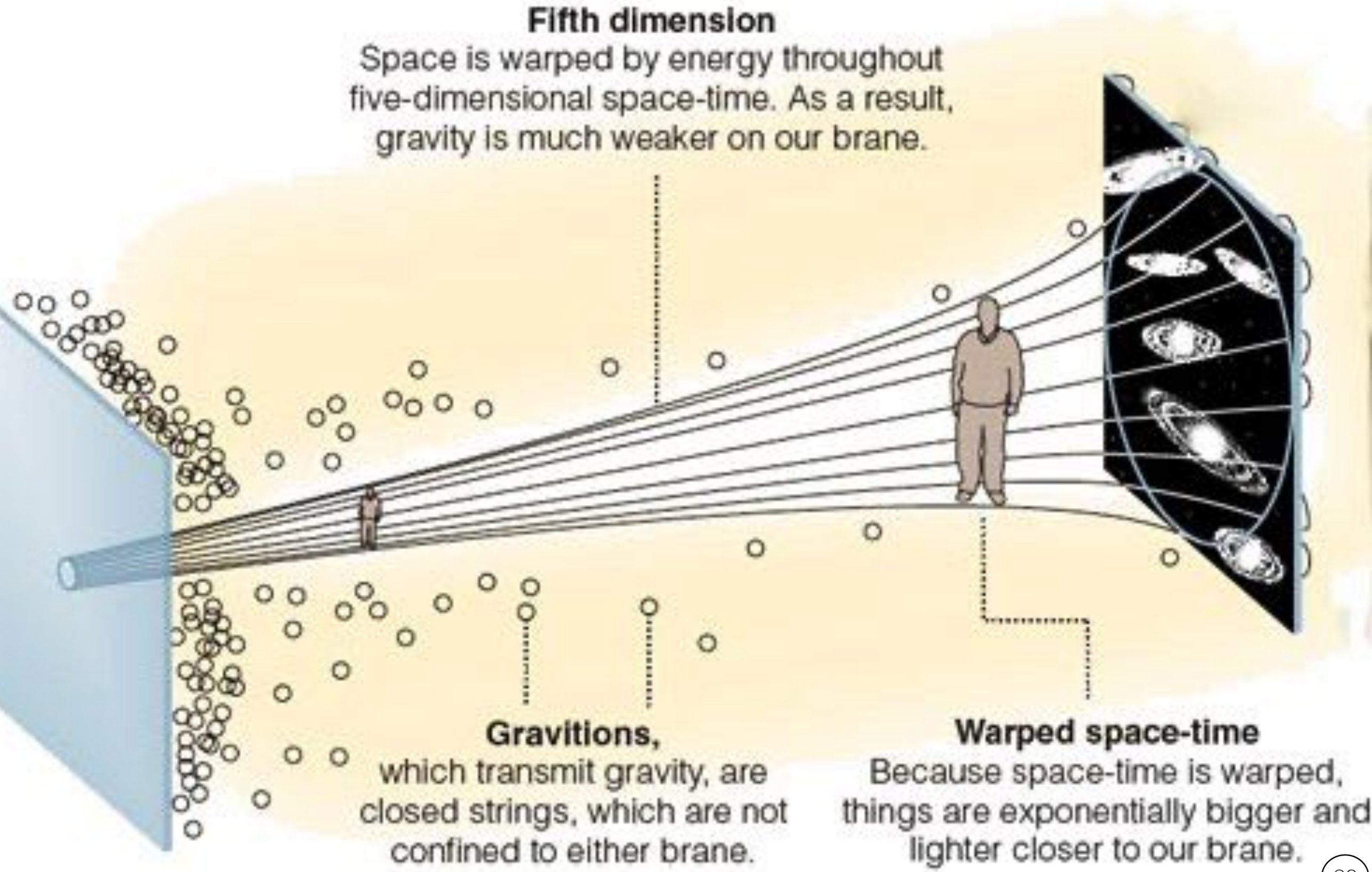
(u) $\pi^0 \rightarrow \gamma + \gamma$

(v) $\Sigma^- \rightarrow n + e + \bar{\nu}_e$

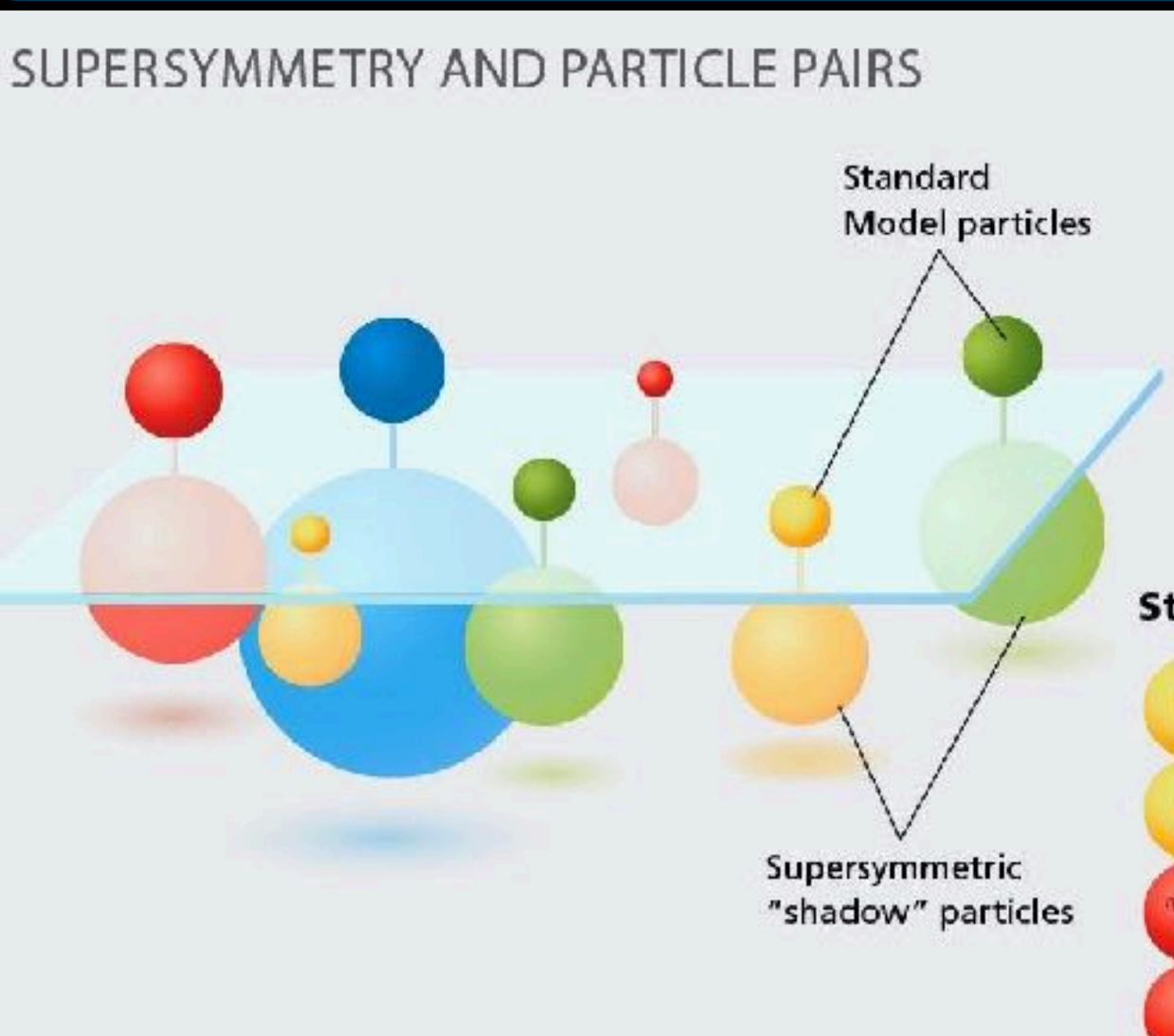
Is that all?

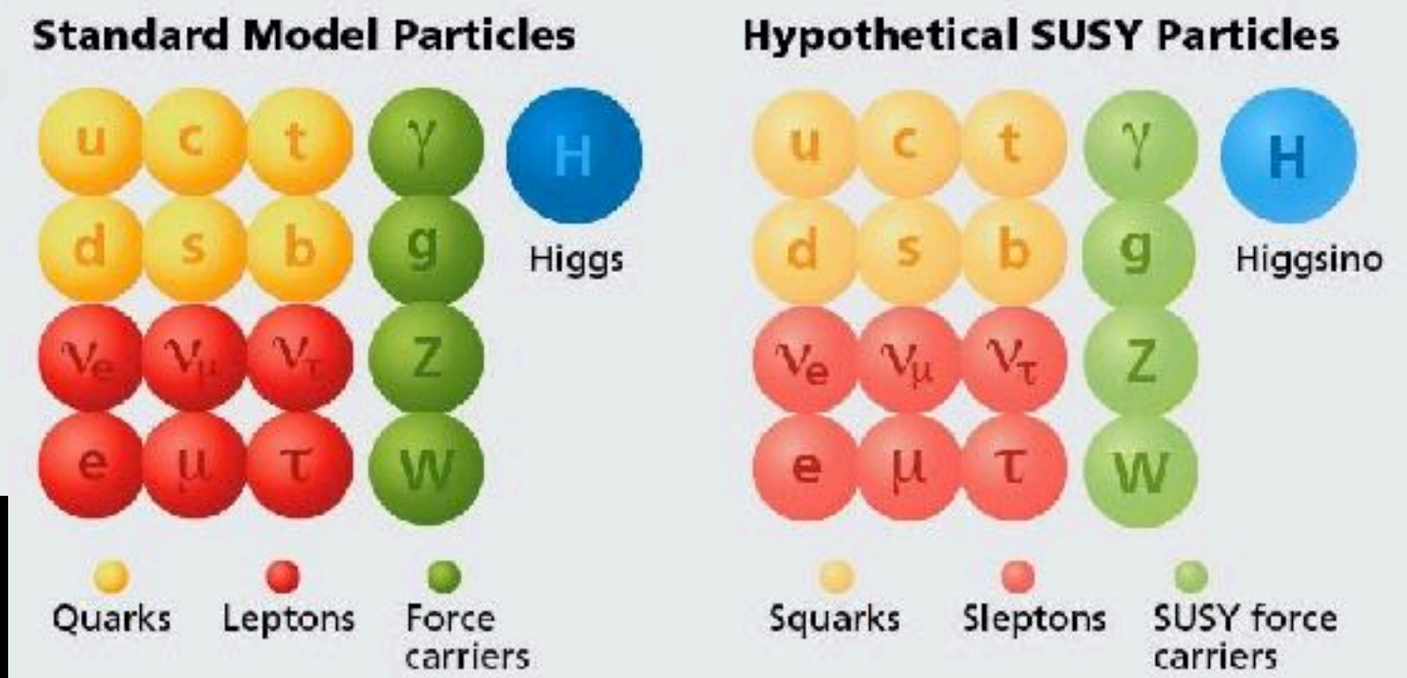

Dark Energy
73%
Dark Matter
23%
Atomic Matter
4.6%
Light
0.005%
Neutrinos
0.0034%

Is that all?


Fifth dimension
Space is warped by energy throughout five-dimensional space-time. As a result, gravity is much weaker on our brane.
Gravitions,
which transmit gravity, are closed strings, which are not confined to either brane.
Warped space-time
Because space-time is warped, things are exponentially bigger and lighter closer to our brane.

Is that all?





Supersymmetry remains just an interesting, if mathematically elegant, hypothesis right now. It would explain why particles have the masses they do and shed light on dark matter. Many versions of the theory suggest the LHC's current experiments will be energetic enough to produce the heavier supersymmetric particles — if they exist.

Where to test?

- Approximately 27km in circumference. Proton beams travel at close to the speed of light to collide.
- The beams travel in opposite directions in separate beam pipes kept at ultrahigh vacuum. They are guided around the accelerator ring by a strong magnetic field maintained by superconducting electromagnets chilled to -271.3°C (1.9K) – a temperature colder than outer space.

Proton + proton collider

The Large Hadron Collider at CERN

Where to test?

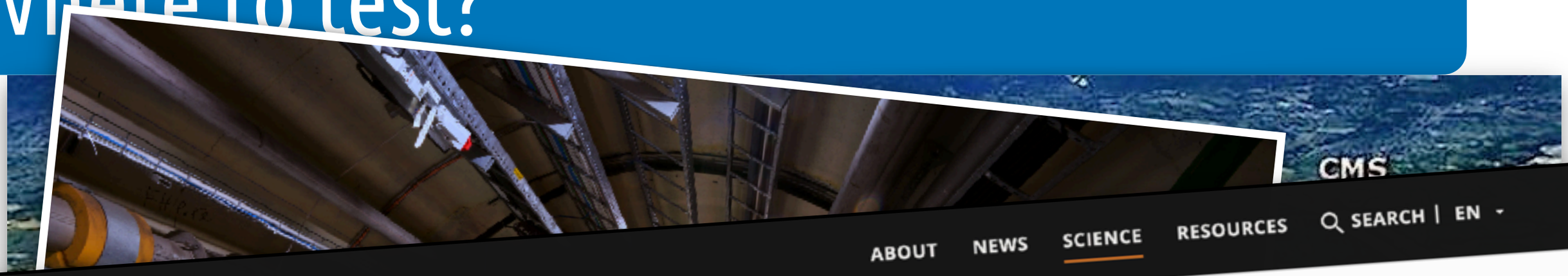
of light to collide.

vacuum. They are
guided around the
perconducting
electromagnets chilled to -271.3°C (1.9K) – a temp
ace.

Proton + proton collider

The Large Hadron Collider at CERN

Where to test?



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A three-in-one vacuum system

The LHC is unusual in that it has three separate vacuum systems: one for the beam pipes, one for insulating the cryogenically cooled magnets and one for insulating the helium distribution line.

To avoid colliding with gas molecules inside the accelerator, the beams of particles in the LHC must travel in a vacuum as empty as interstellar space. In the cryomagnets and the helium distribution line, the vacuum serves a different purpose. Here, it acts as a thermal insulator, to reduce the amount of heat that seeps from the surrounding room-temperature environment into the cryogenic parts which are kept at at 1.9 K (-271.3°C).

The largest vacuum system in the world

With a total of 104 kilometres of piping under vacuum, the vacuum system of the LHC is among the largest in the world. The insulating vacuum, equivalent to some 10^{-6} mbar, is made up of an impressive 50 km of piping, with a combined volume of 15,000 cubic metres, more than enough to fill the nave of a cathedral. Building this vacuum system required more than 250,000 welded joints and 18,000 vacuum seals. The remaining 54 km of pipes under vacuum are the beam pipes, through which the LHC's two beams travel. The pressure in these pipes is in the order of 10^{-10} to 10^{-11} mbar, a vacuum almost as rarefied as that found on the surface of the Moon. The LHC's vacuum systems are fitted with 170 Bayard-Alpert ionisation gauges and 1084 Pirani and Penning gauges to monitor the vacuum pressure.

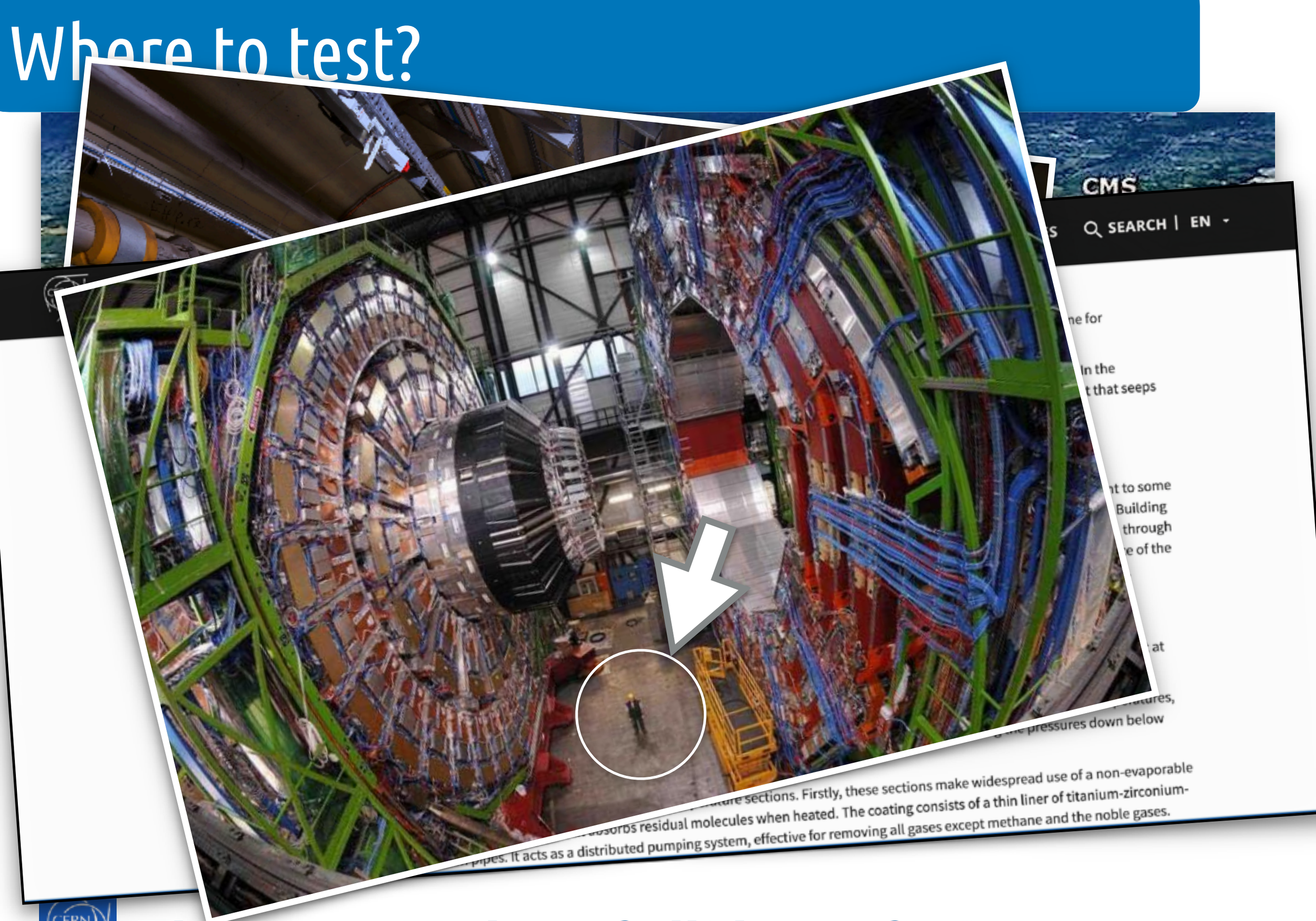
A vacuum thinner than the interstellar void

Ultra-high vacuum is needed for the pipes in which particle beams travel. This includes 48 km of arc sections, kept at 1.9 K, and 6 km of straight sections, kept at room temperature, where beam-control systems and the insertion regions for the experiments are located.

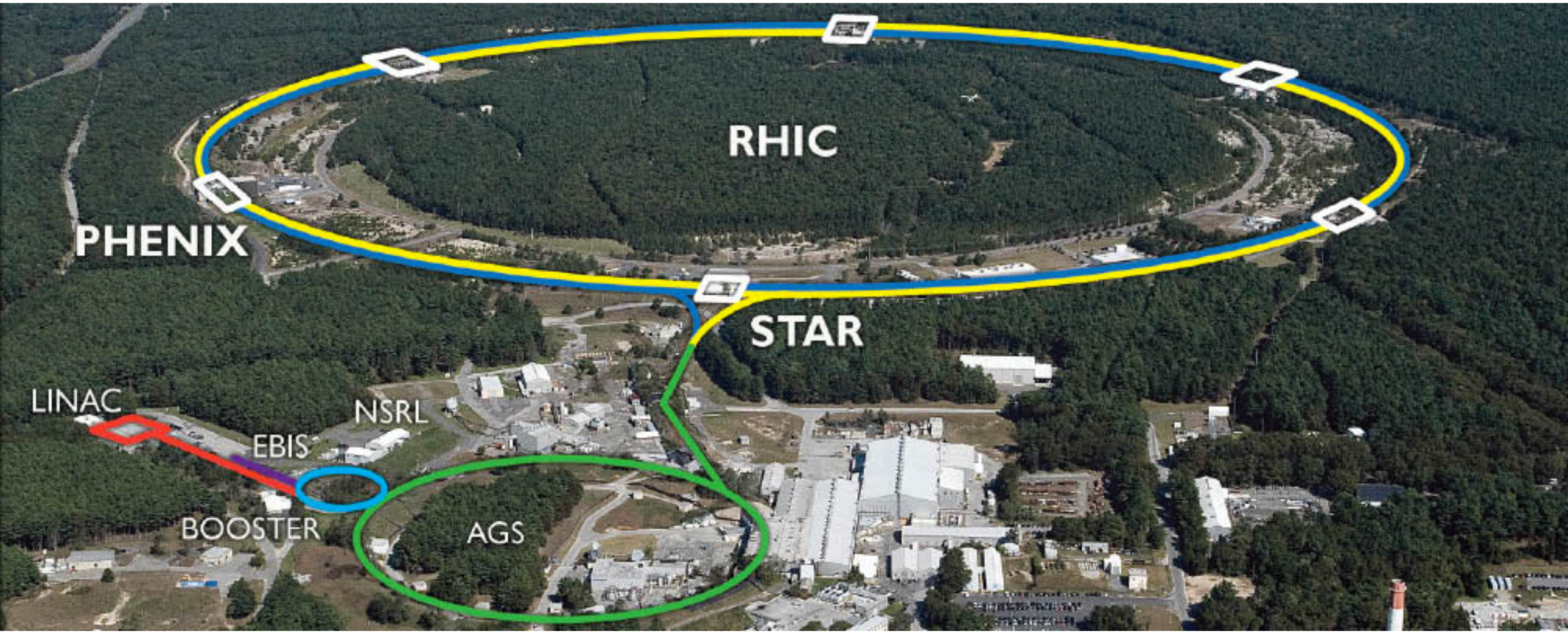
In the arcs, the ultra-high vacuum is maintained by cryogenic pumping of 9000 cubic metres of gas. As the beam pipes are cooled to extremely low temperatures, the gases condense and adhere to the walls of the beam pipe by adsorption. Just under two weeks of pumping are required to bring the pressures down below 1.013×10^{-10} mbar (or 10^{-13} atmospheres).

Two important design features maintain the ultra-high vacuum in the room-temperature sections. Firstly, these sections make widespread use of a non-evaporable "getter coating" – developed and industrialized at CERN – that absorbs residual molecules when heated. The coating consists of a thin liner of titanium-zirconium-vanadium alloy deposited inside the beam pipes. It acts as a distributed pumping system, effective for removing all gases except methane and the noble gases.

Where to test?



Where to test?



Gold + polarised proton collider

Brookhaven National Laboratory

Relativistic Heavy Ion Collider

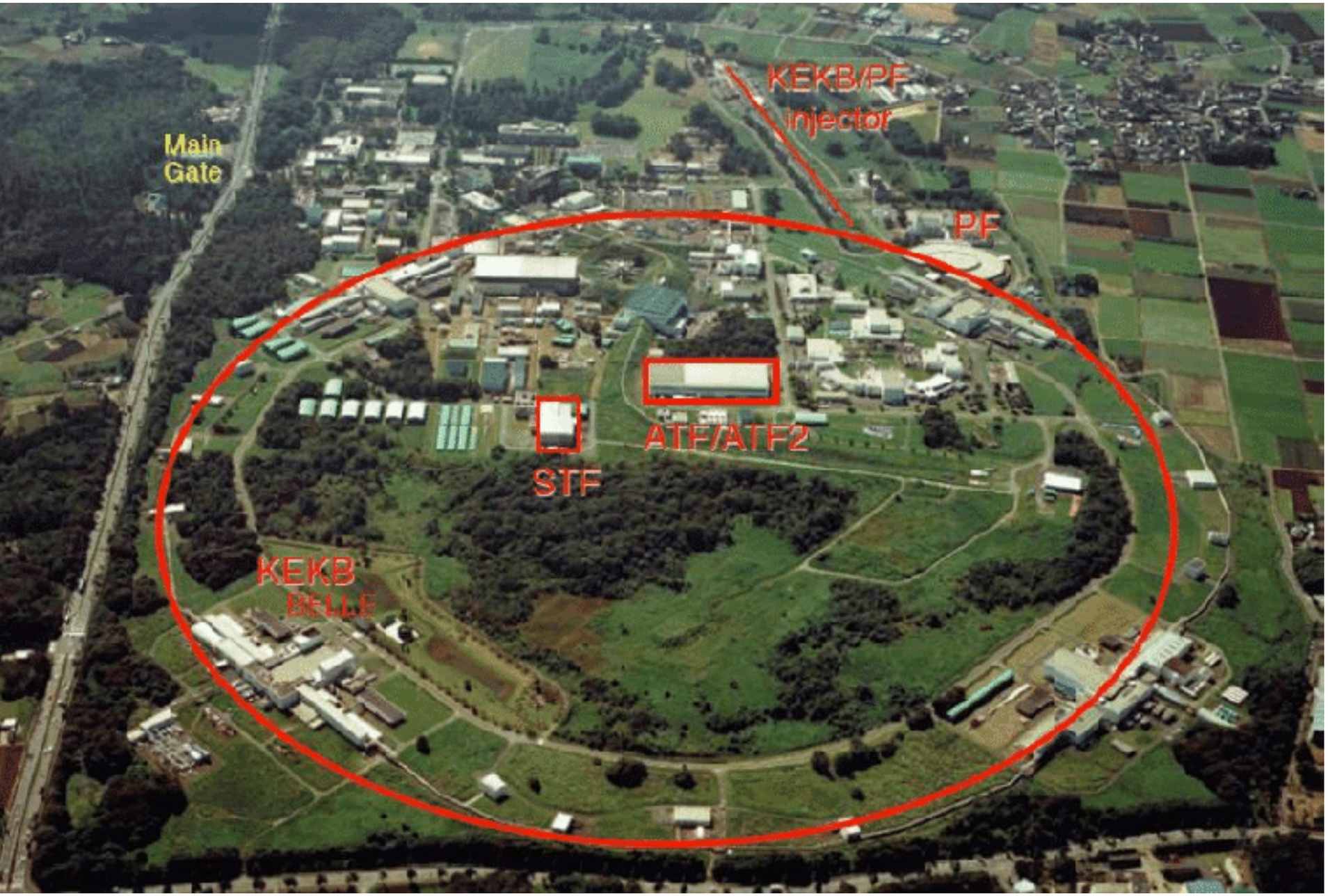
Where to test?

Proton + anti-proton collider

Fermilab

Tevatron

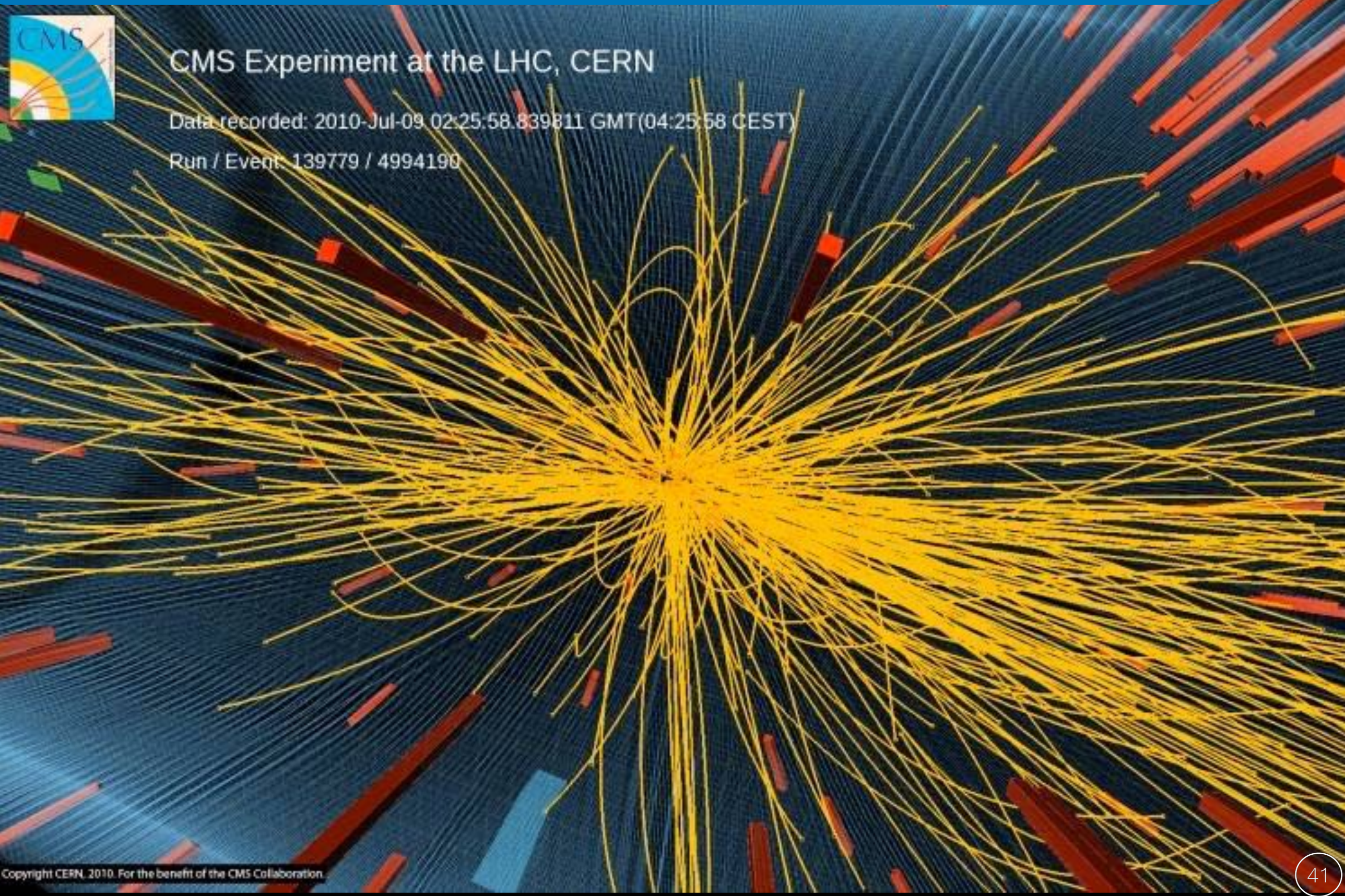
Where to test?



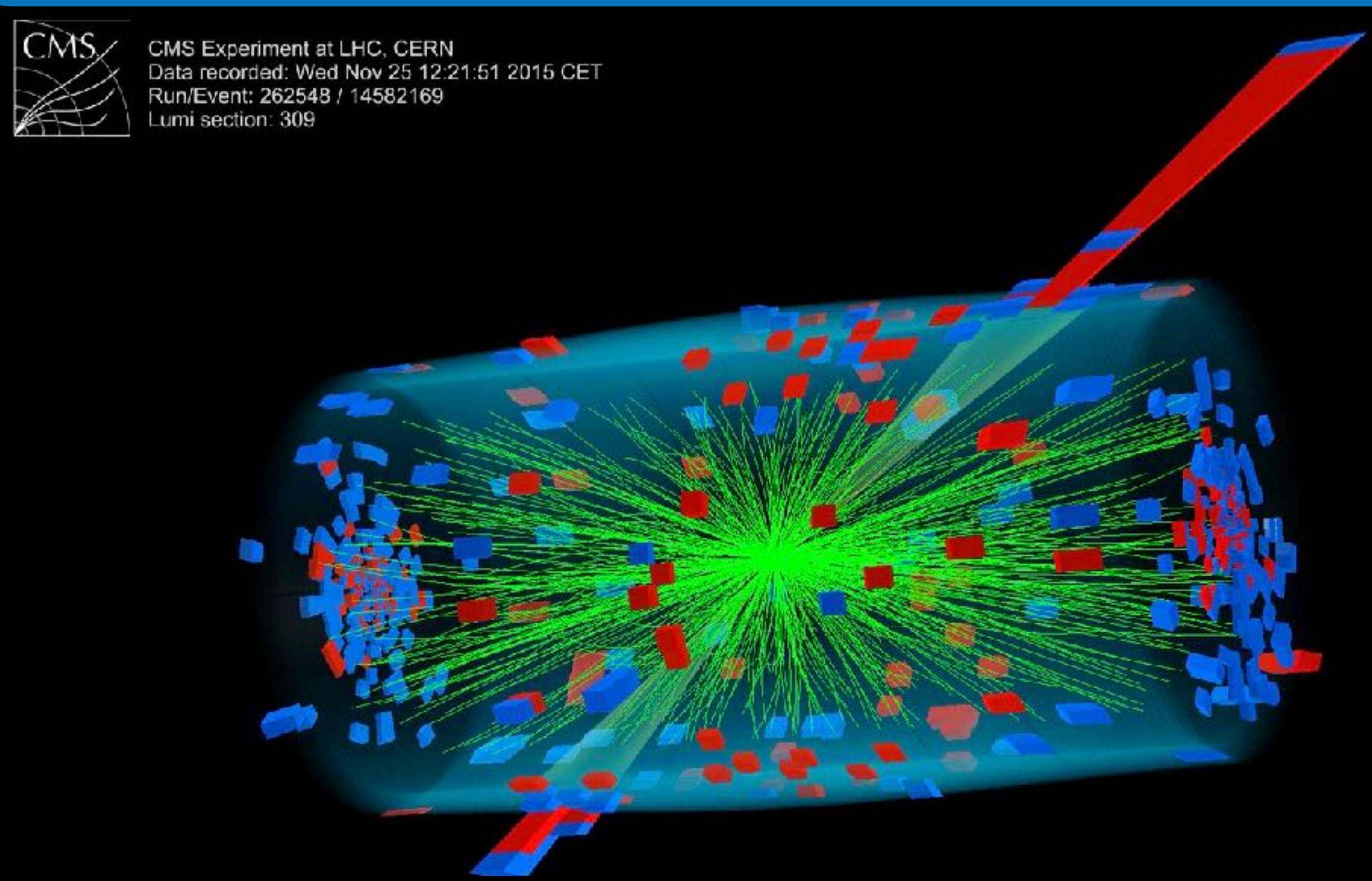
KEK-JAPAN

Neutrino Experiments

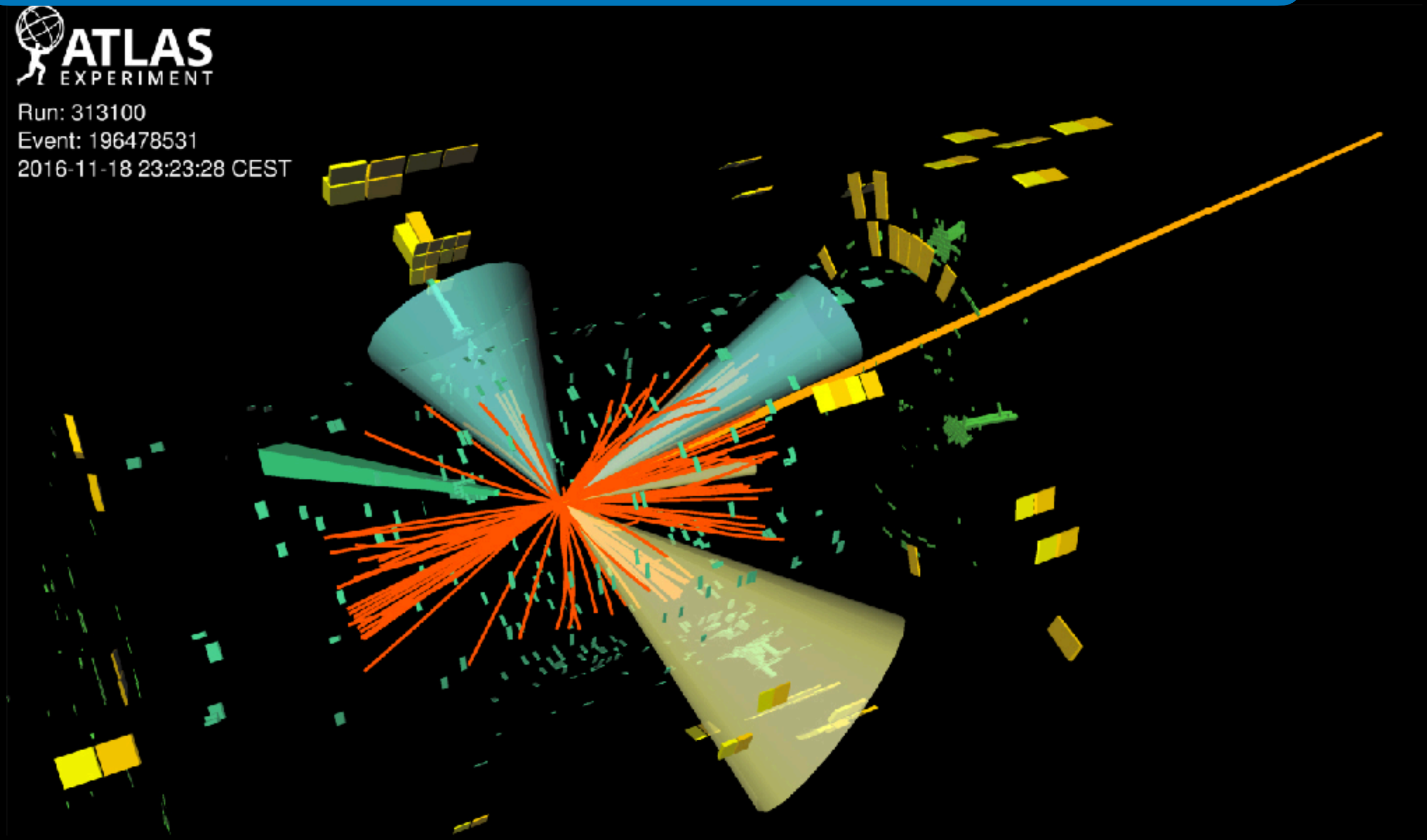
Inside a collider



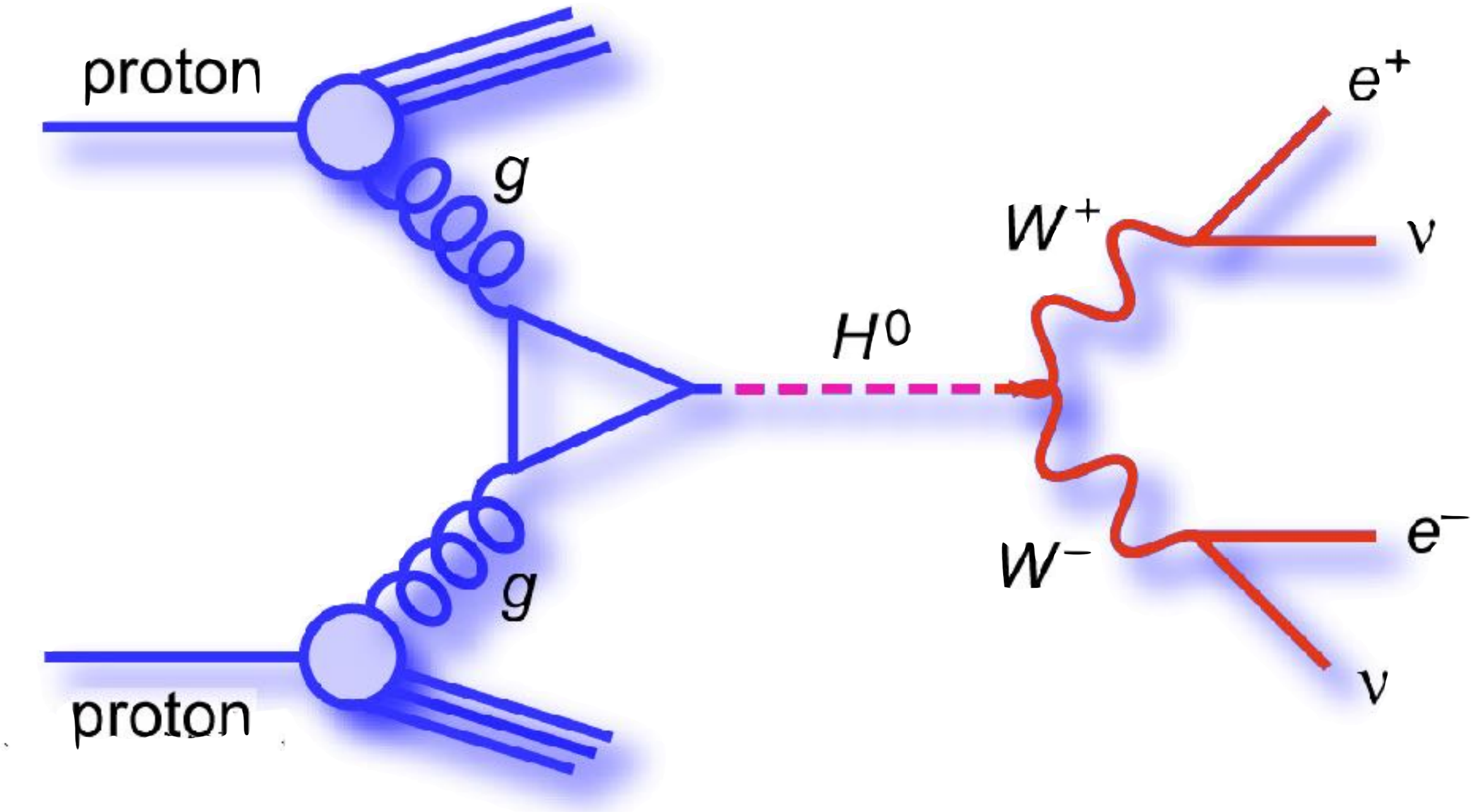
Inside a collider



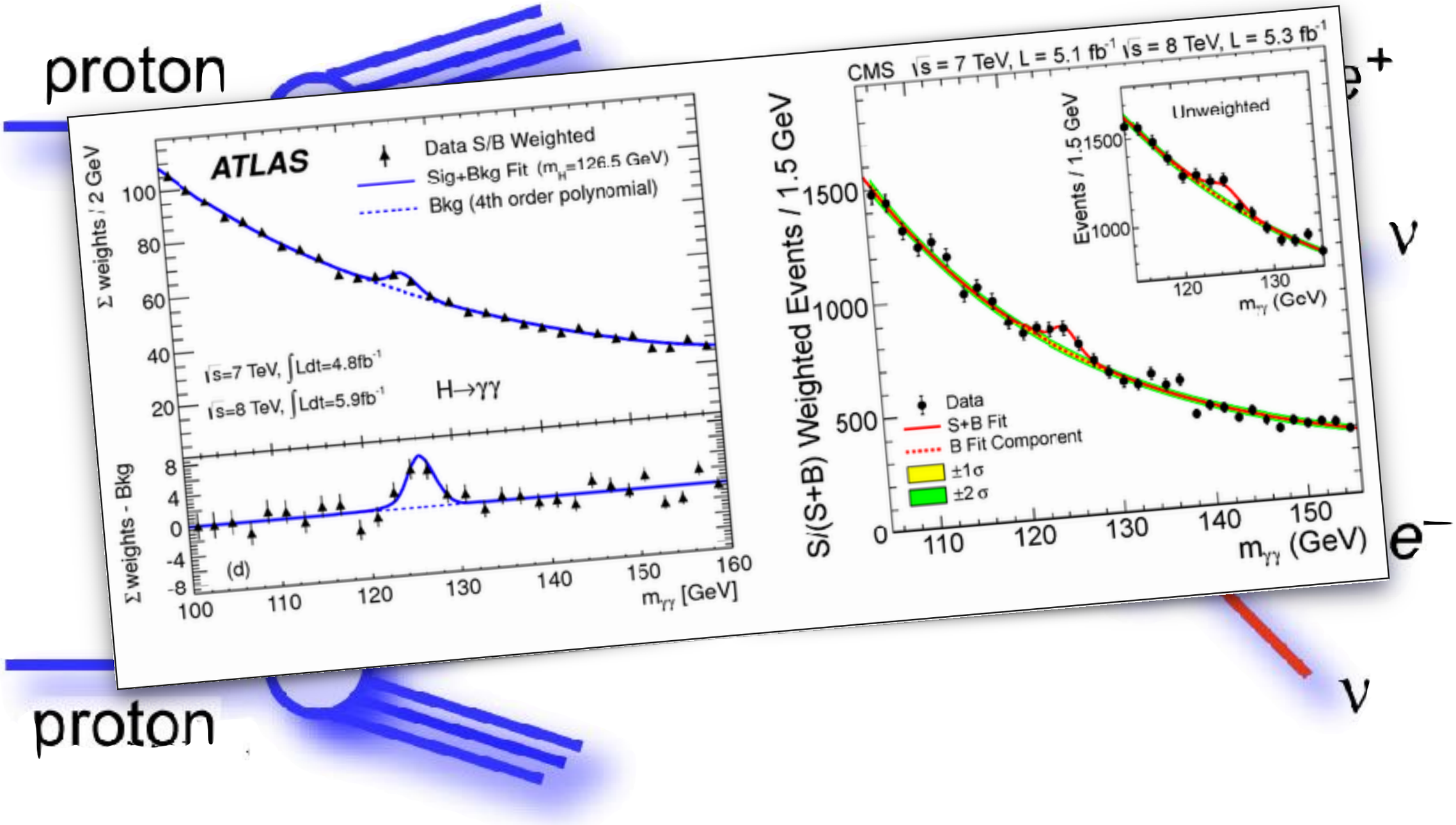
Inside a collider



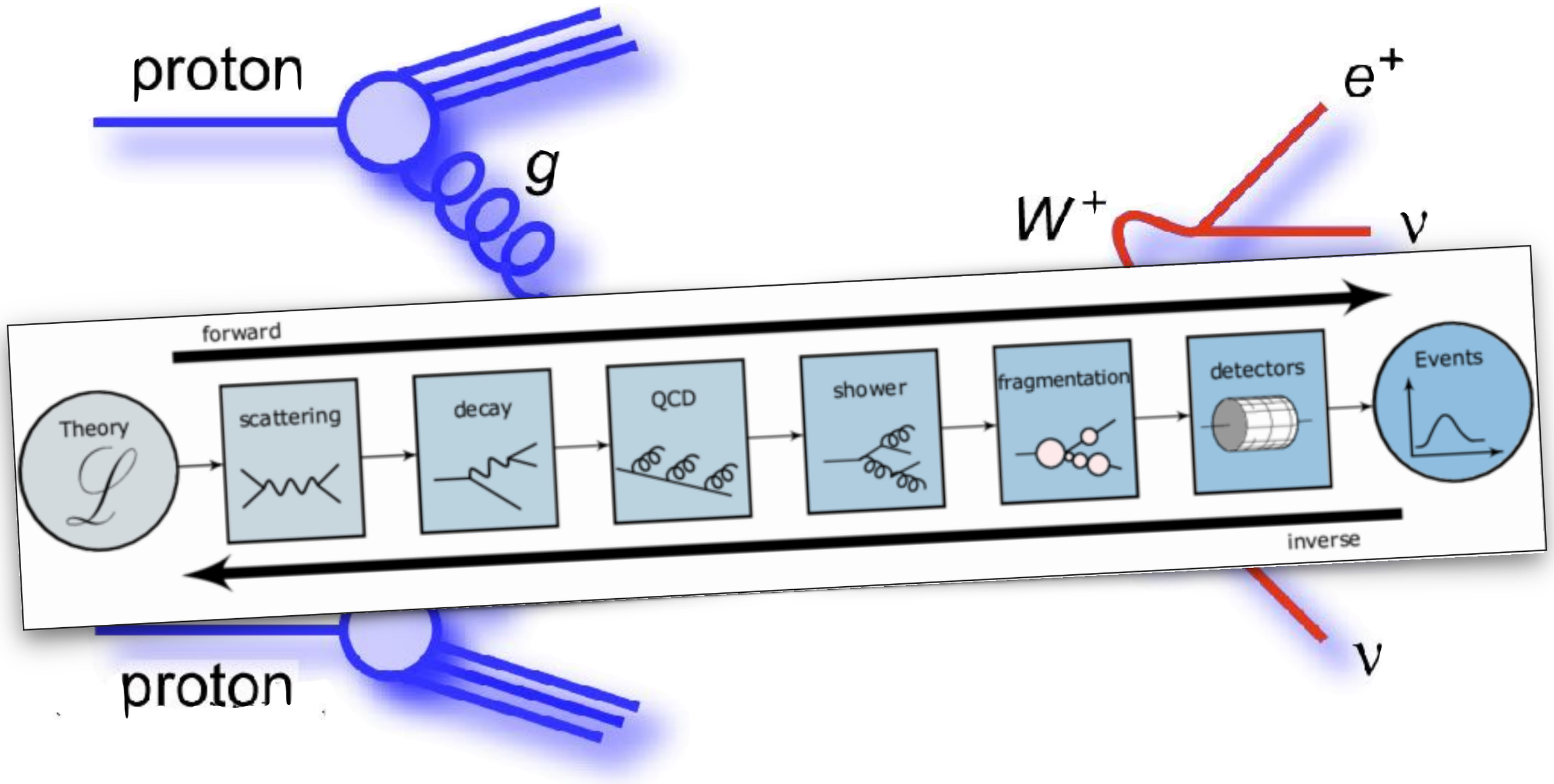
How to see?



How to see?


proton
proton
ATLAS
Data S/B Weighted
Sig+Bkg Fit (m_H=126.5 GeV)
Bkg (4th order polynomial)
Σ weights / 2 GeV
√s=7 TeV, ∫Ldt=4.8fb⁻¹
√s=8 TeV, ∫Ldt=5.9fb⁻¹
H→γγ
Σ weights - Bkg
(d)
$m_{\gamma\gamma}$ [GeV]
CMS √s = 7 TeV, L = 5.1 fb⁻¹ √s = 8 TeV, L = 5.3 fb⁻¹
Unweighted
Events / 1.5 GeV
$m_{\gamma\gamma}$ (GeV)
Data
S+B Fit
B Fit Component
±1σ
±2σ
S/(S+B) Weighted Events / 1.5 GeV
$m_{\gamma\gamma}$ (GeV)
e⁺
e⁻
ν
ν

How to see?


proton
g
e^+
W^+
ν
forward
Theory
$\mathscr{L}$
scattering
decay
QCD
shower
fragmentation
detectors
Events
inverse
proton
ν